

PCS–P125 power

Rev A0 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PCS–P125: two–level SiC bridge (6 x SG2M040170HJ per switch), split DC link, LCL, ports

01 Control connectors, defaults

PCS_PC (= PC pin for pin, 2x32) and PCS_X (2x8) to PV–CTL (PCS assembly); command pull–downs, RDY pull–up, DC– star points

File: sheets/01_pc.kicad_sch

02 DC link, bleeders, sensing

5 + 5 x C3D1U147 in two series halves, midpoint M (to the C_f star); bleeder per half; VB = V(DC+), VA = V(M) around VMID (REF3030E)

File: sheets/02_dclink.kicad_sch

03 Phase leg a, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section a); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/03_lega.kicad_sch

04 Phase leg b, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section b); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/04_legb.kicad_sch

05 Phase leg c, power stage

2 x 6 x SG2M040170HJ on Al2O3 pads (earthed section c); 10 k + 1 nF per gate; 3 sub–cells: 3 x 2.2 uF film + RC damper each

File: sheets/05_legc.kicad_sch

06 Leg a high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/06_gdah.kicad_sch

07 Leg a low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/07_gdal.kicad_sch

08 Leg b high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/08_gdbh.kicad_sch

09 Leg b low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/09_gdbl.kicad_sch

10 Leg c high–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT

File: sheets/10_gdch.kicad_sch

11 Leg c low–side gate drive

gdrv.channel() 6 x SG2M040170HJ; NSI6651ASC–Q1 +18/–3.5 V, 8.75/7.5 ohm per device; PNP turn–off buffers, Miller clamps, booster, DESAT; bias SN6505B + T_BIAS4

File: sheets/11_gdcl.kicad_sch

12 Phase a LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_A: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/12_filta.kicad_sch

13 Phase b LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_B: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/13_filtb.kicad_sch

14 Phase c LCL filter, current sensor

Switch node -> L1 (chassis) -> sensor -> C_f node F_C: C_f 2 x 25 uF + R_d–C_d to the star M (DC midpoint) -> L2 (chassis) -> AC port

File: sheets/14_filtc.kicad_sch

15 AC voltage sensing

6 x (6 x 1 M ARHV06 + 4.99 k to VMID, 4.7 nF): VC1–3 at the C_f nodes, VG1–3 at the AC terminal taps; OPA2388 followers to PCS_X

File: sheets/15_acsense.kicad_sch

16 Port B power path (battery)

Varistor Y + monitor, aR fuses

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

CAUTIONED: BUS–meas and life–sign–in–work / aC–sensors very diverse, do not build (outputs PCS + PWR, test–rel–elect–kV, valve–f–a–re–a–t–i–o–n–d–r–e–s–i–p–t–s–design/pcs_spec/pcs_spec.json; 200 A time, Y capacitors, board entries, removable PE link, port status to PCS_PC, insulation monitor (port.lean_imd)

Domains: LIVE (DC– = GND = AGND, the AC side included), GD_<leg><H/L> gate islands (functional: NSI6651, T_BIAS4), PE (pads, heatsink, NTC lugs, Y / varistor / IMD crossings), SELV (AUX–T1).

Stages 1–6 drawn (gen/pcs_power.py header); OPEN items listed there and in the design check.

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

CAUTIONED: BUS–meas and life–sign–in–work / aC–sensors very diverse, do not build (outputs PCS + PWR, test–rel–elect–kV, valve–f–a–re–a–t–i–o–n–d–r–e–s–i–p–t–s–design/pcs_spec/pcs_spec.json; 200 A time, Y capacitors, board entries, removable PE link, port status to PCS_PC, insulation monitor (port.lean_imd)

Domains: LIVE (DC– = GND = AGND, the AC side included), GD_<leg><H/L> gate islands (functional: NSI6651, T_BIAS4), PE (pads, heatsink, NTC lugs, Y / varistor / IMD crossings), SELV (AUX–T1).

Stages 1–6 drawn (gen/pcs_power.py header); OPEN items listed there and in the design check.

File: sheets/18_leanB_drive.kicad_sch

19 DC terminals, insulation test

DC terminals 200 A time, Y capacitors, board entries, removable PE link, port status to PCS_PC, insulation monitor (port.lean_imd)

File: sheets/19_dcterm.kicad_sch

20 AC port: contactors, SPD, RCM

L2 out -> K1 -> K2 -> CM choke -> AC terminals; coil drivers (fail–safe high side); 3 x TVT25751 + 3 x Y1 to PE_T; type–B RCM sensor

File: sheets/20_acport.kicad_sch

21 PE bonds, heatsink, NTC inputs

Board PE bond, heatsink sections (earthed), heatsink NTC lug probes, L1 winding NTC leads, NTC filters (biased on the control board)

File: sheets/21_earthbond.kicad_sch

22 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/22_selvconnector.kicad_sch

23 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = DC–; 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uF / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/23_aux75_input.kicad_sch

24 AUX75 start–up, brown–in, OV, timer

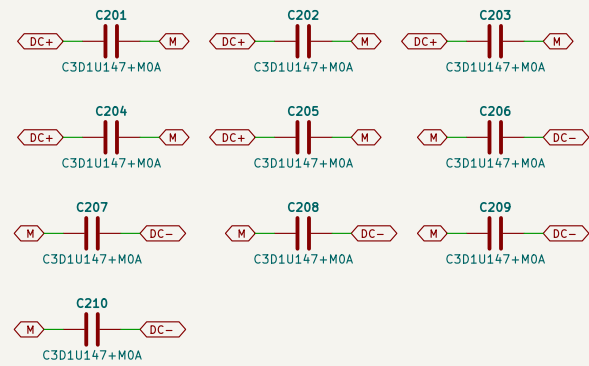
As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 35–80 ms

File: sheets/24_aux75_start.kicad_sch

Not bench–validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC–; gate islands functional; HV–PE basic

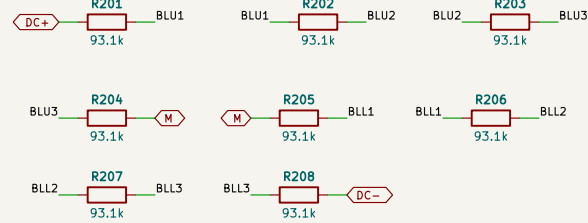
Sheet: 1
File: F:\pcs\revC1\revC1.kicad_sch
Title: PCS–P125 power
Size: 1000 x 1000
KiCad: 7.0.0

SPLIT DC LINK



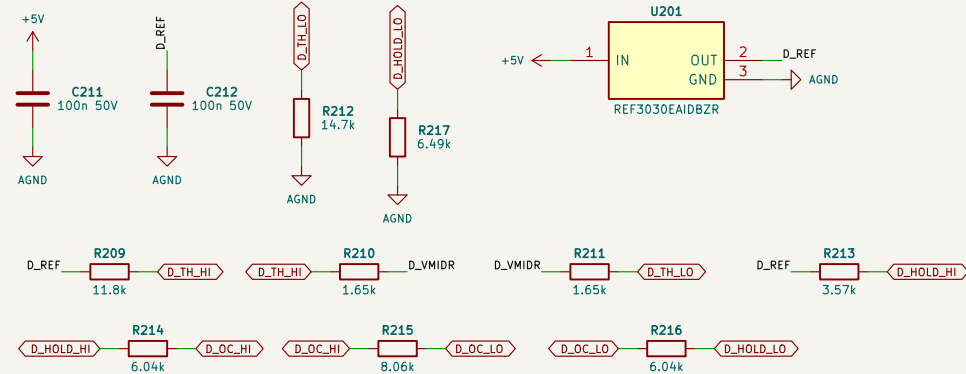
Upper half DC+ to M, lower half M to DC-, 5 x 140 uF each (700 uF per half, 350 uF in series).
M is the C_f star point of the LCL filter (no midpoint control: two-level).

BLEEDERS (BALANCE THE HALVES, DISCHARGE THE LINK)



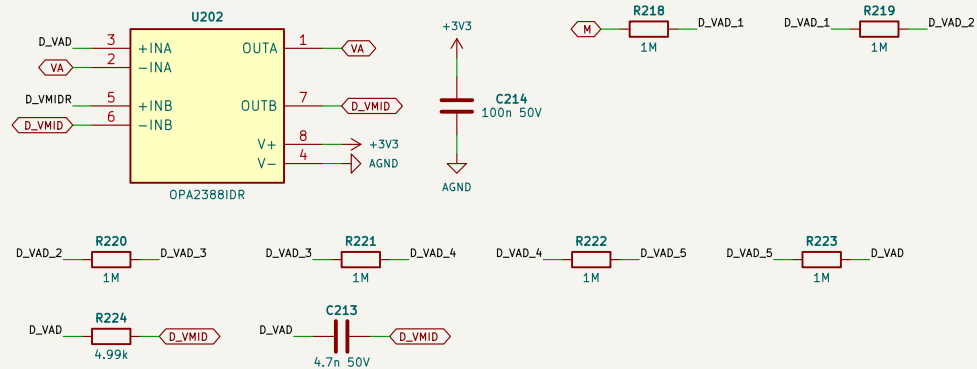
4 x 93.1k 1210 per half (372.4k); label 'wait 15 min'.

REFERENCE (PORT.LEAN_REFS: REF3030E, VMID 1.646 V AND THE DC-PORT THRESHOLD LADDERS)



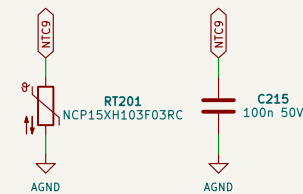
Shared with the DC port (stage 4). VMID buffered here; IA, IA_H, VAX of PCS_PC sit on it.

MIDPOINT DIVIDER (6 X 1 M ARHV06 + 4.99 K TO VMID, 4.7 NF: 1 V = 1203 V)



VA = VMID + V(M)/1203 (OPA2388 follower); its second half is the VMID follower. VB (DC+) comes from the DC port's bank divider (lean_dividers).

DC-LINK TEMPERATURE (NTC9 TO PCS_X)



Murata NCP15 10 k on the board between the film capacitors; biased on the control board.

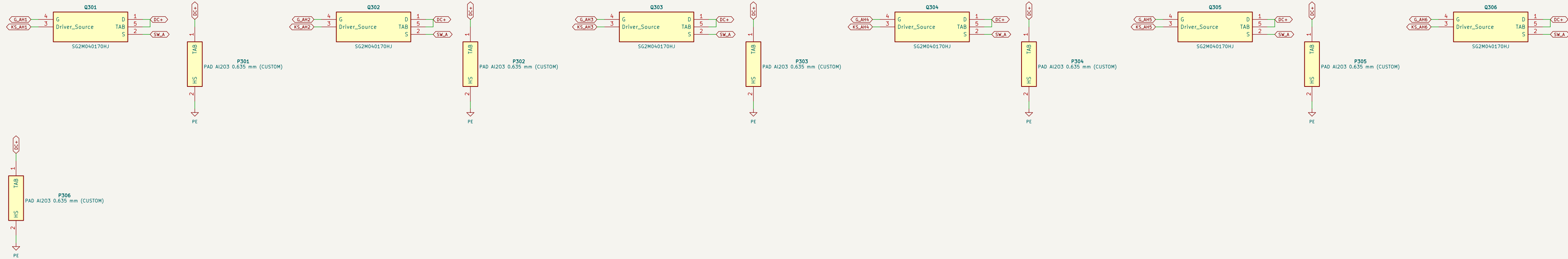
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 3
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /02 DC link, bleeders, sensing/
File: 02_dclink.kicad_sch

Title: PCS-P125 power – DC link, bleeders, sensing

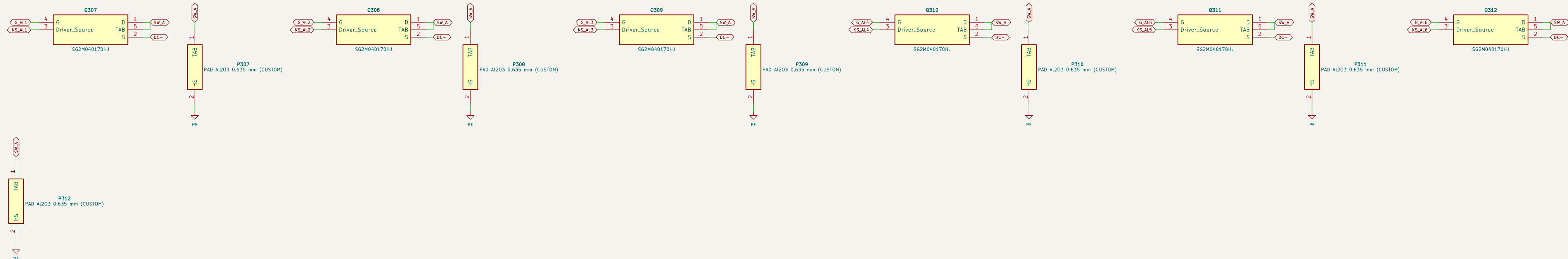
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 3/27

LEG A HIGH SWITCH: 6 X SG2M040170HJ



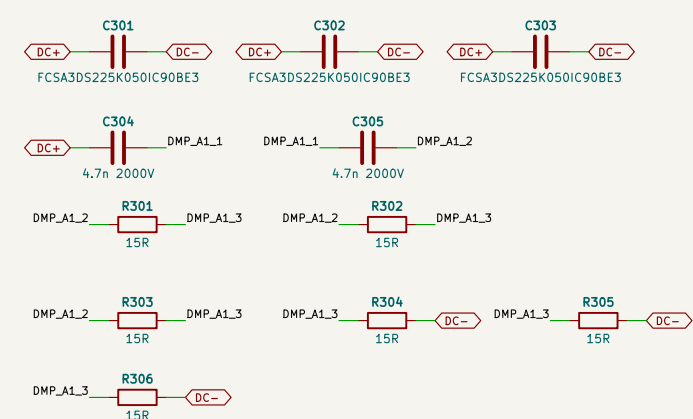
Per device: own gate net G_AH<k> and Kelvin net KS_AH<k> (gate network on the gate-drive sheet). BOM NOTE: R_DS(on) of the six within +/- 5 % (V_GS 18 V, 38 A, 25 C), V_GS(th) within +/- 0.25 V, one lot per switch (pcs_spec device_acceptance, PCM-06).

LEG A LOW SWITCH: 6 X SG2M040170HJ



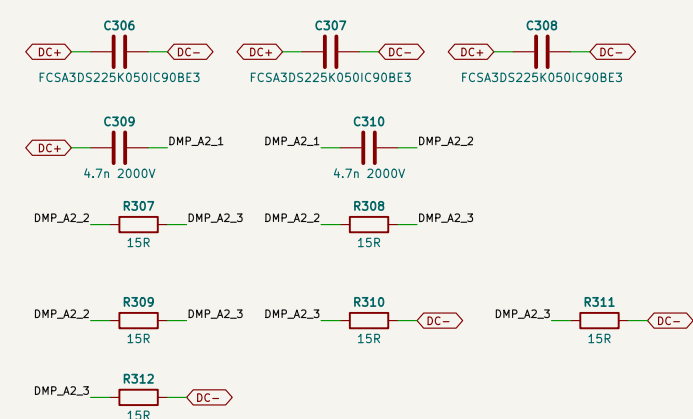
Per device: own gate net G_AL<k> and Kelvin net KS_AL<k> (gate network on the gate-drive sheet). BOM NOTE: R_DS(on) of the six within $\pm 5\%$ (V_{GS} 18 V, 38 A, 25 C), $V_{GS(th)}$ within ± 0.25 V, one lot per switch (pcs_spec device_acceptance, PCM-06).

LEG A SUB-CELL 1: DECOUPLING + RC DAMPER



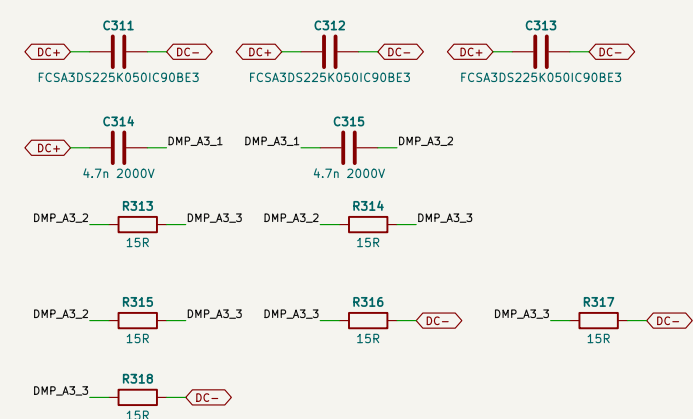
3 x FCSA3DS225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series.
loop at the pins <= 3 nH (LAYOUT).

LEG A SUB-CELL 2: DECOUPLING + RC DAMPER



3 x FCSA3DS225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG A SUB-CELL 3: DECOUPLING + RC DAMPER



3 x FCSA3DS225 at devices 5-6 (high and low): damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series loop at the pins ≤ 3 nH (LAYOUT).

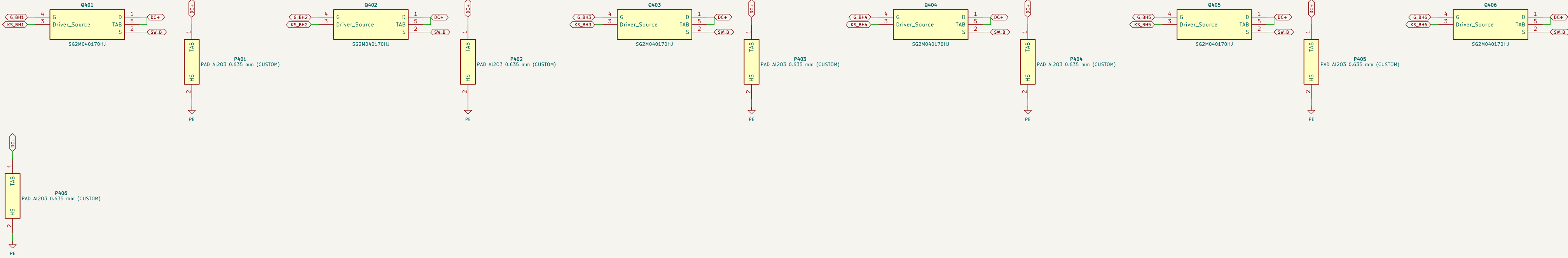
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 4
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /03 Phase leg a, power stage/
File: 03_lega.kicad_sch

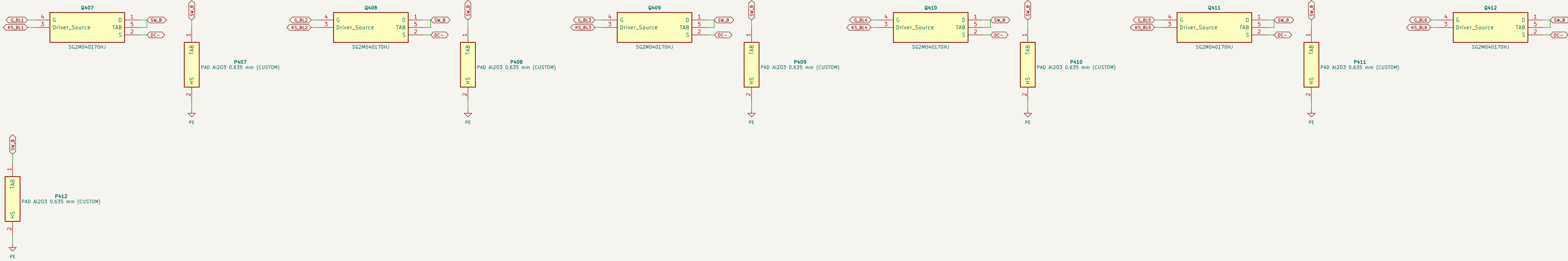
Title: PCS-P125 power – Phase leg a, power stage

Size: A2	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 4/27

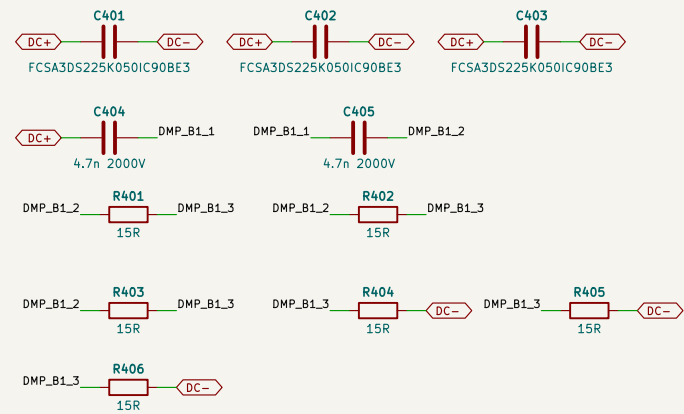
LEG B HIGH SWITCH: 6 X SG2M040170HJ



LEG B LOW SWITCH: 6 X SG2M040170HJ

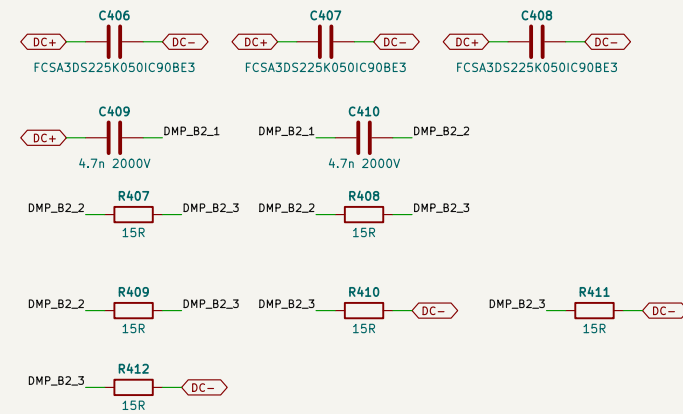


LEG B SUB-CELL 1: DECOUPLING + RC DAMPER



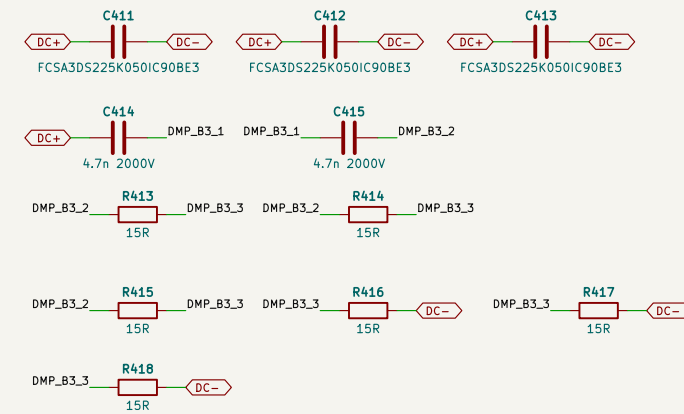
3 x FCSA3DS225 at devices 1-2 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG B SUB-CELL 2: DECOUPLING + RC DAMPER



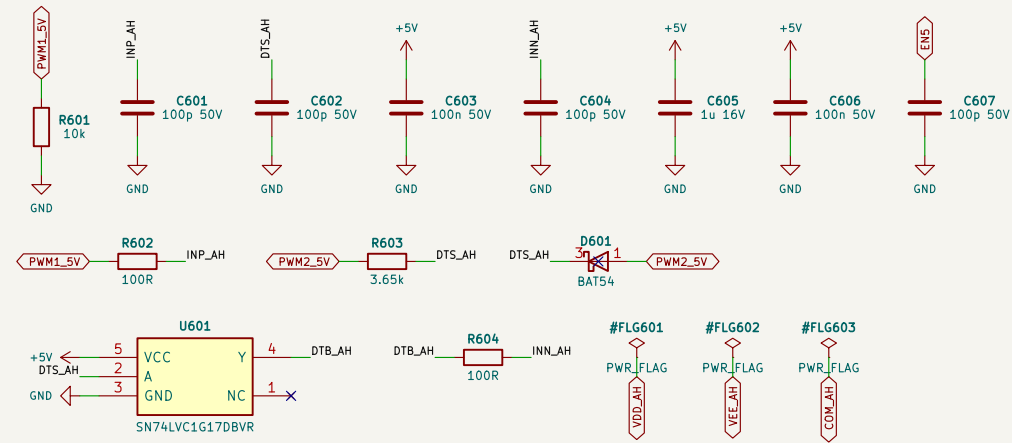
3 x FCSA3DS225 at devices 3-4 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

LEG B SUB-CELL 3: DECOUPLING + RC DAMPER



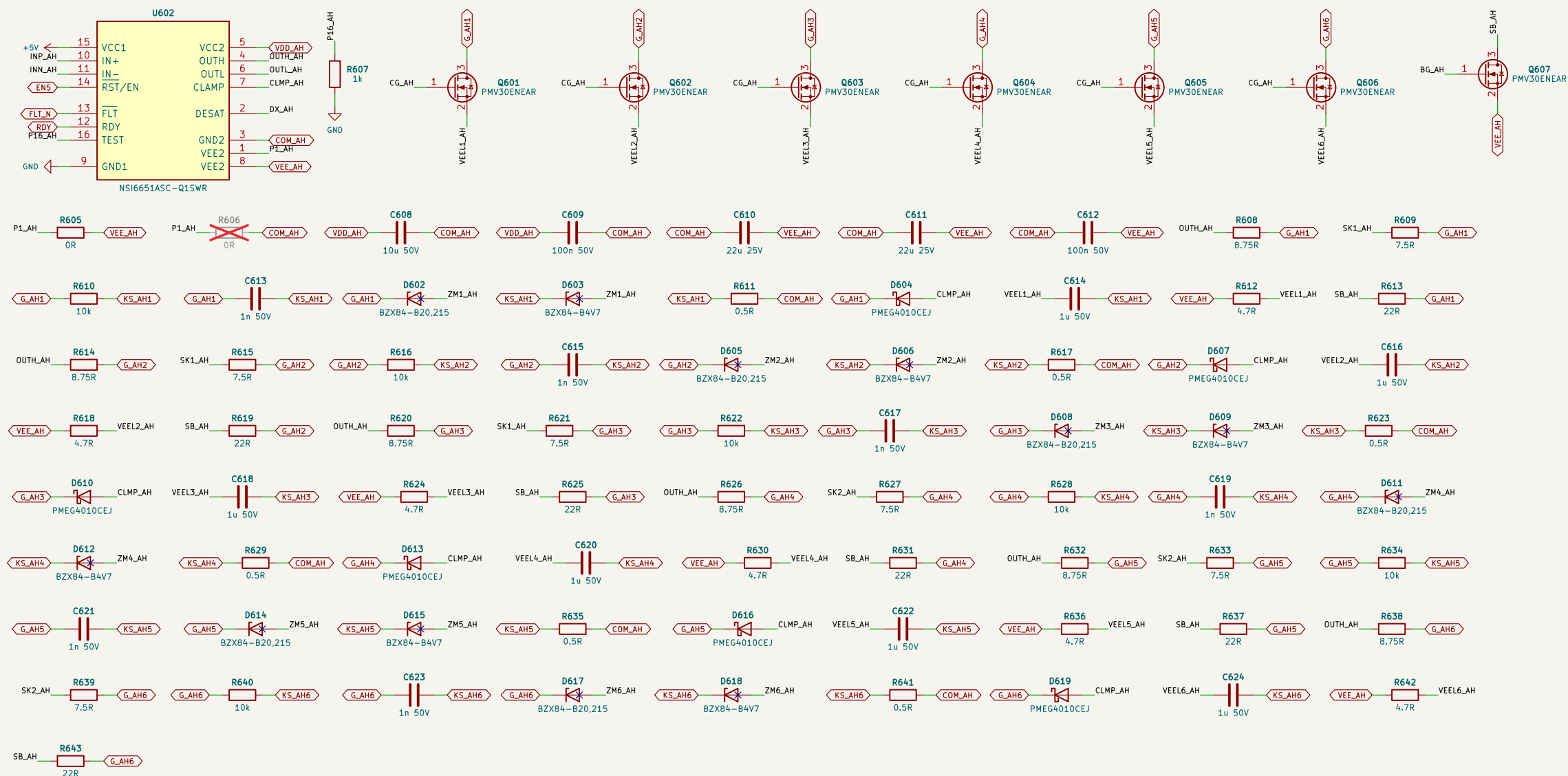
3 x FCSA3DS225 at devices 5-6 (high and low); damper 2 x (3 x 15R) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH (LAYOUT).

AH: LOGIC SIDE



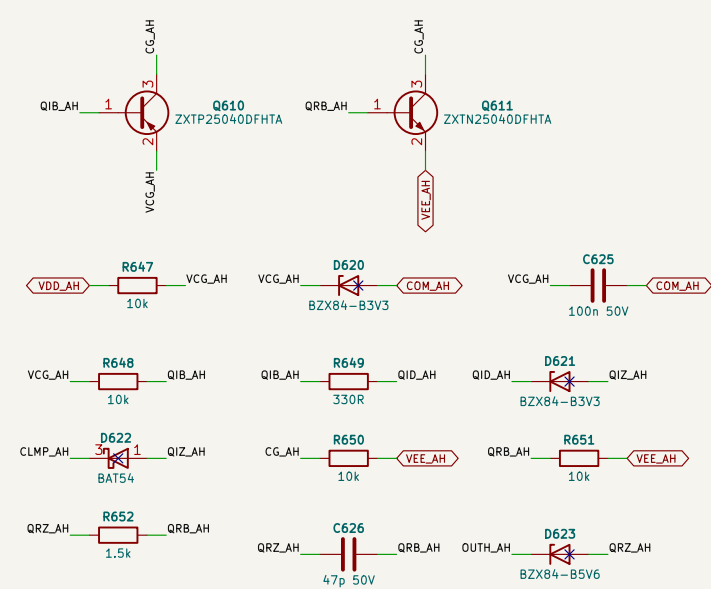
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

AH: DRIVER AND 6 GATES



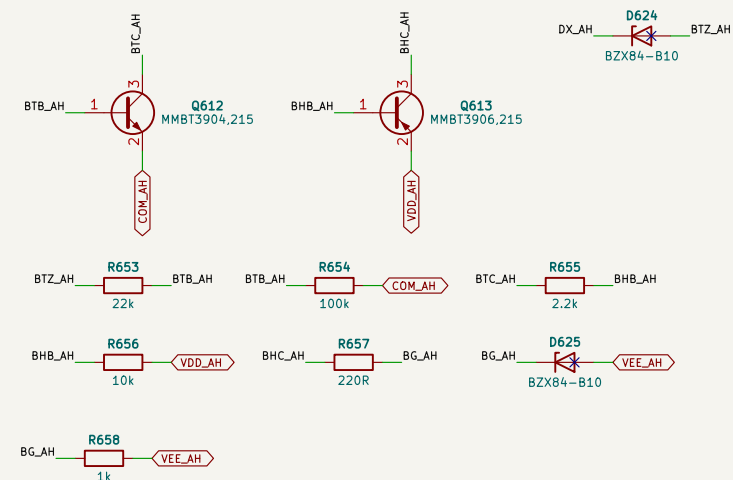
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

AH: MILLER CLAMP CONTROL



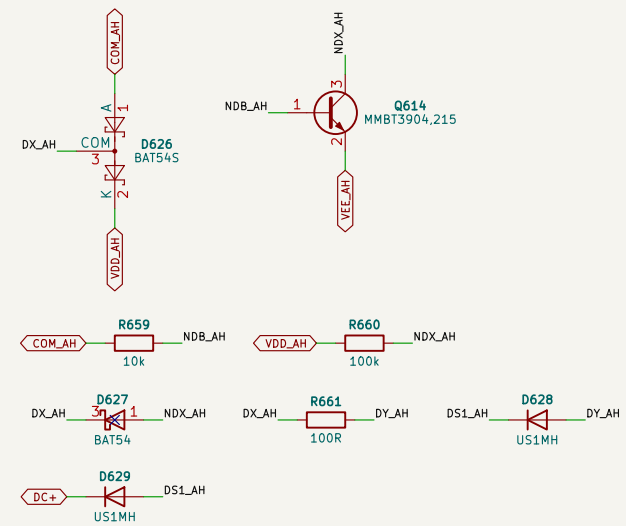
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

AH: SHORT-CIRCUIT BOOSTER



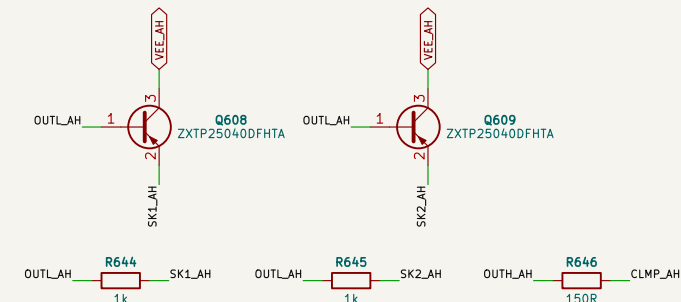
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

AH: DESAT



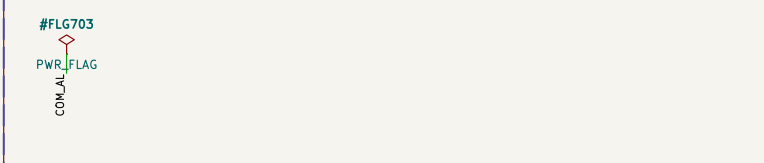
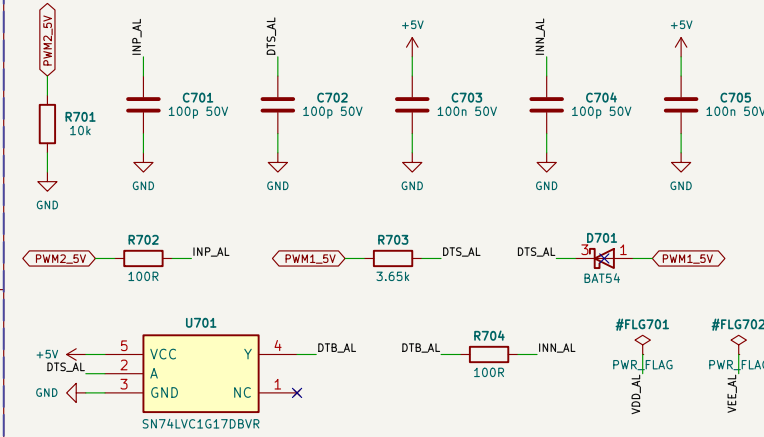
DESAT pin (Internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

AH: TURN-OFF BUFFER



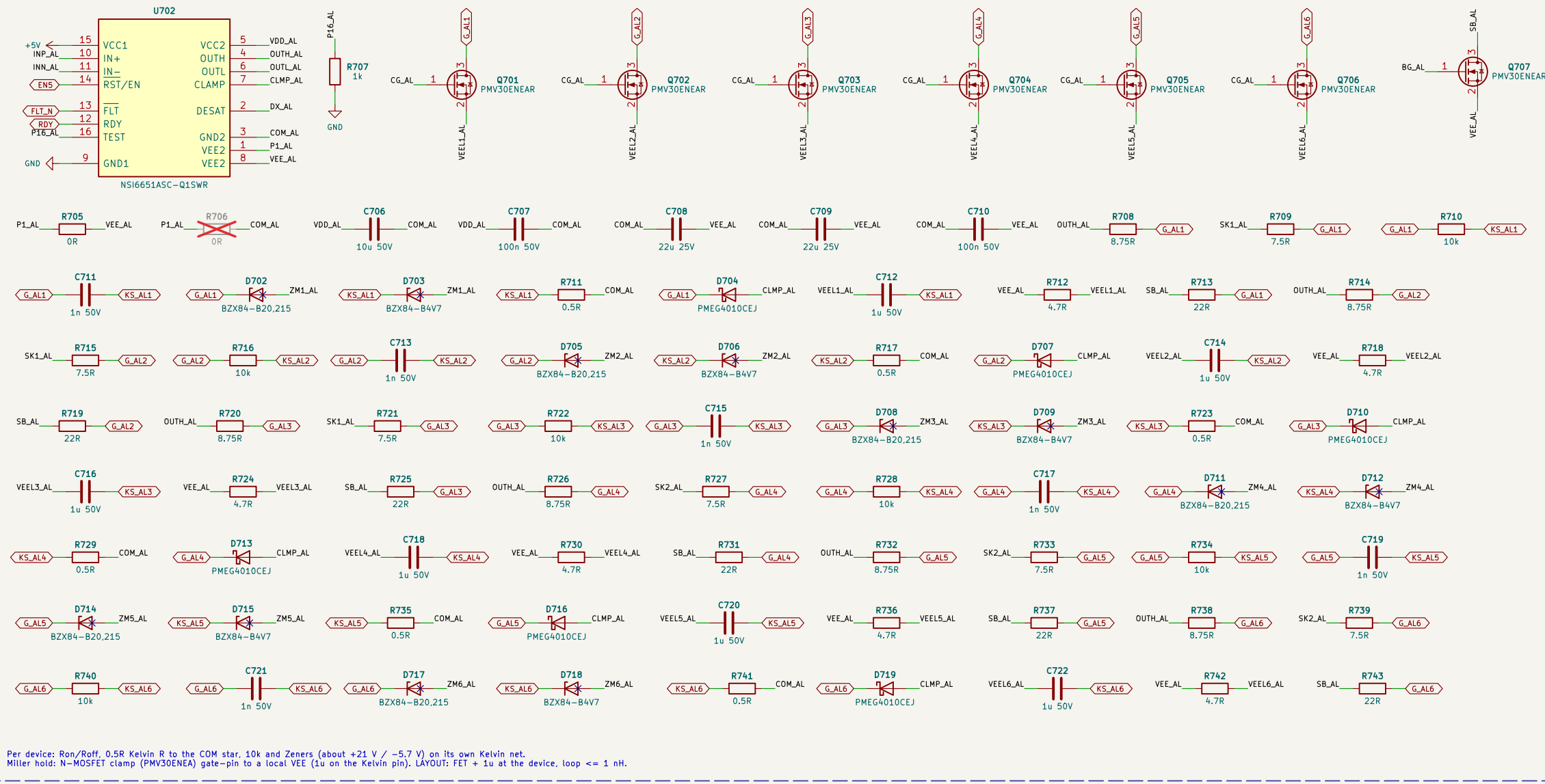
2 x XZTP25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

AL: LOGIC SIDE



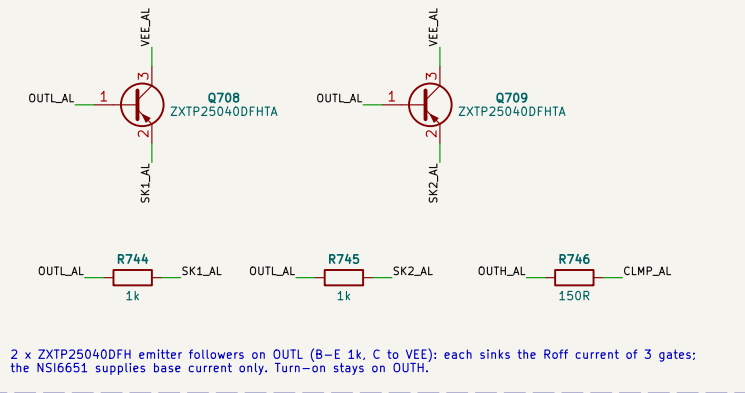
IN+ = own PWM; IN- = complementary switch, falling edge detected 3.6V_{th}/100p; (C701C5 = 10k dead time = deadTime_{max} + 100p); DESAT pin (active low) pulls RDT low.

AL: DRIVER AND 6 GATES



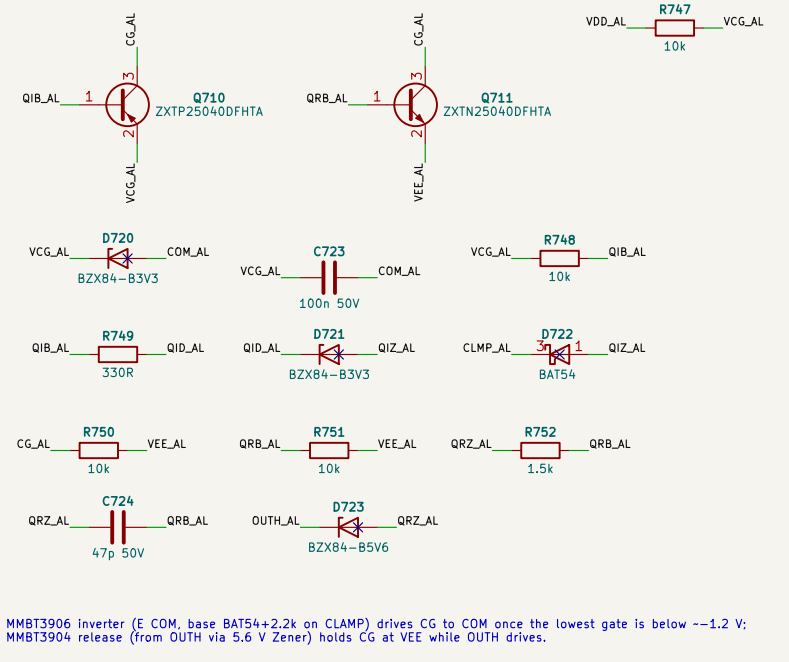
Per device: Rdt/Rdft: 0.58 Kelvin; R to the COM star: 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller mode to MOSFET clamp (PHEG401DCEJ) gate-pin to a base VEE (to on the Kelvin pin) LATCH: ET = 1u at the device, loop <= 1 ns.

AL: TURN-OFF BUFFER



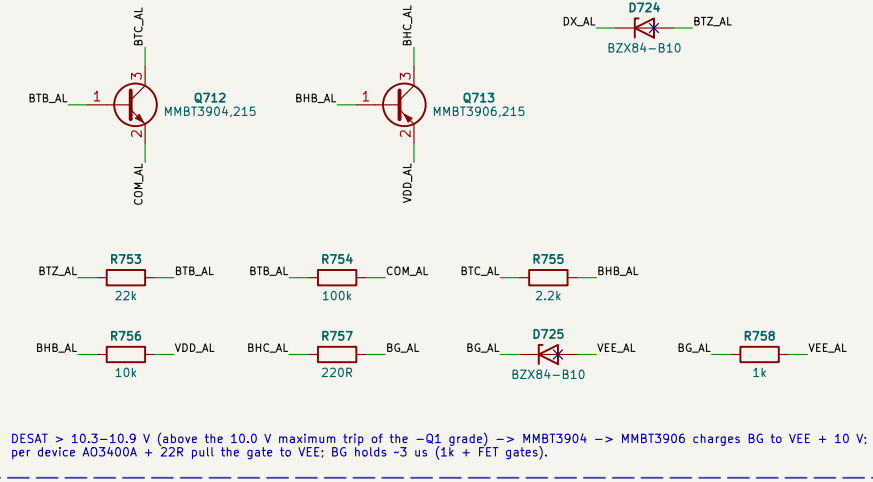
2 x ZXTF25040DPH1x another follows on OUTL (B-E is C to VEE) each sinks the Rdt current of 3 gates; the MOSFETs supply base current only, turn-on delay on OUTL.

AL: MILLER CLAMP CONTROL



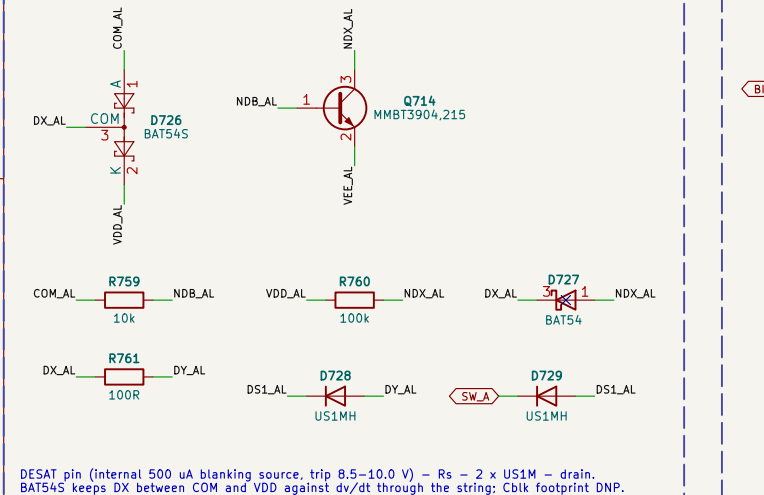
MHBT3906: Inverter (C COM, base B8204-8205 on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MHBT3906: release (from OUTP via 5.6 V Zener) holds CG at VEE while OUTP drives.

AL: SHORT-CIRCUIT BOOSTER



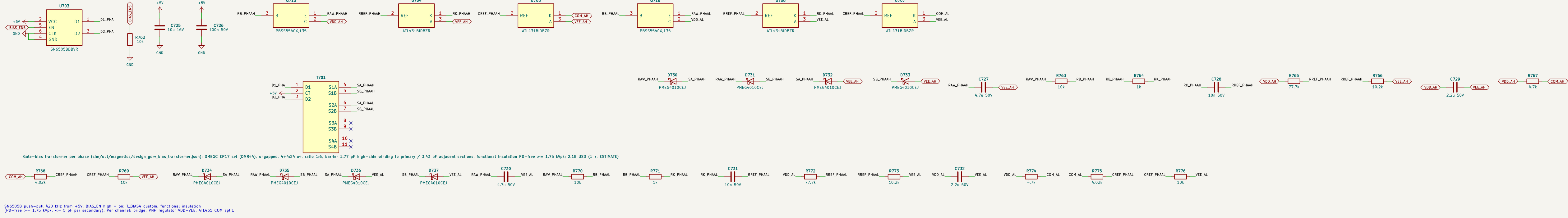
DESAT = 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MHBT3904 -> MHBT3906 charges BG to VEE + 10 V per device A03404 + 22M pull the gate to VEE; BG holds -3 on (14 + FET gates).

AL: DESAT



DESAT pin (Interval 500 uA blanking source, trip 8.5-10.0 V) - R_{dt} = 2 x USM - drain; B820G keeps BG between COM and VEE against Rv/G through the wiring; C5B: Redundant ONP.

PHA: GATE BIAS (PER PHASE)



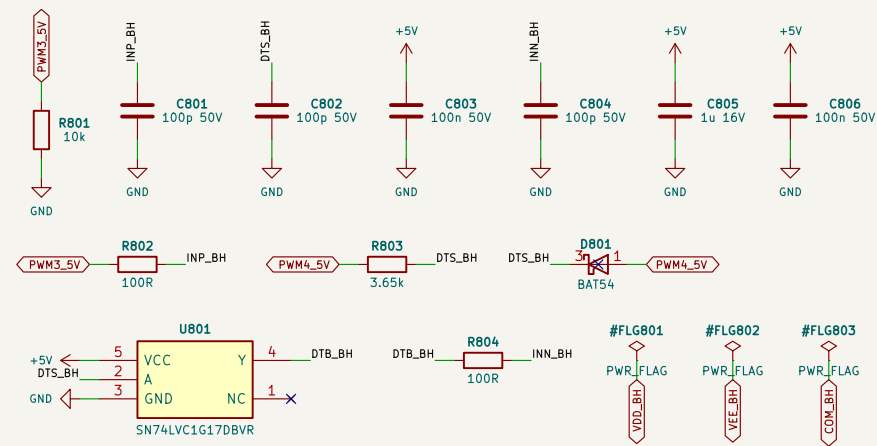
Gate-bias transformer per phase (sim/out/magnetics/design_gdnv_bias_transformer.json): DMSC EP17 set (DMR4), ungapped, 4+424 vA, ratio 1:6, barrier 1.77 pF of high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVp; 2.18 USD (± k, ESTIMATE)

S80500B push-pull A20 kVt from +5V, BAS_EN high = on T_BIAS custom, functional insulation (PD-free >= 1.75 kVp, <= 5 pF per secondary); Per channel bridge: PNP regulator VDD-VEE, ATL431 COM split.

LEG A: LAYOUT RULES

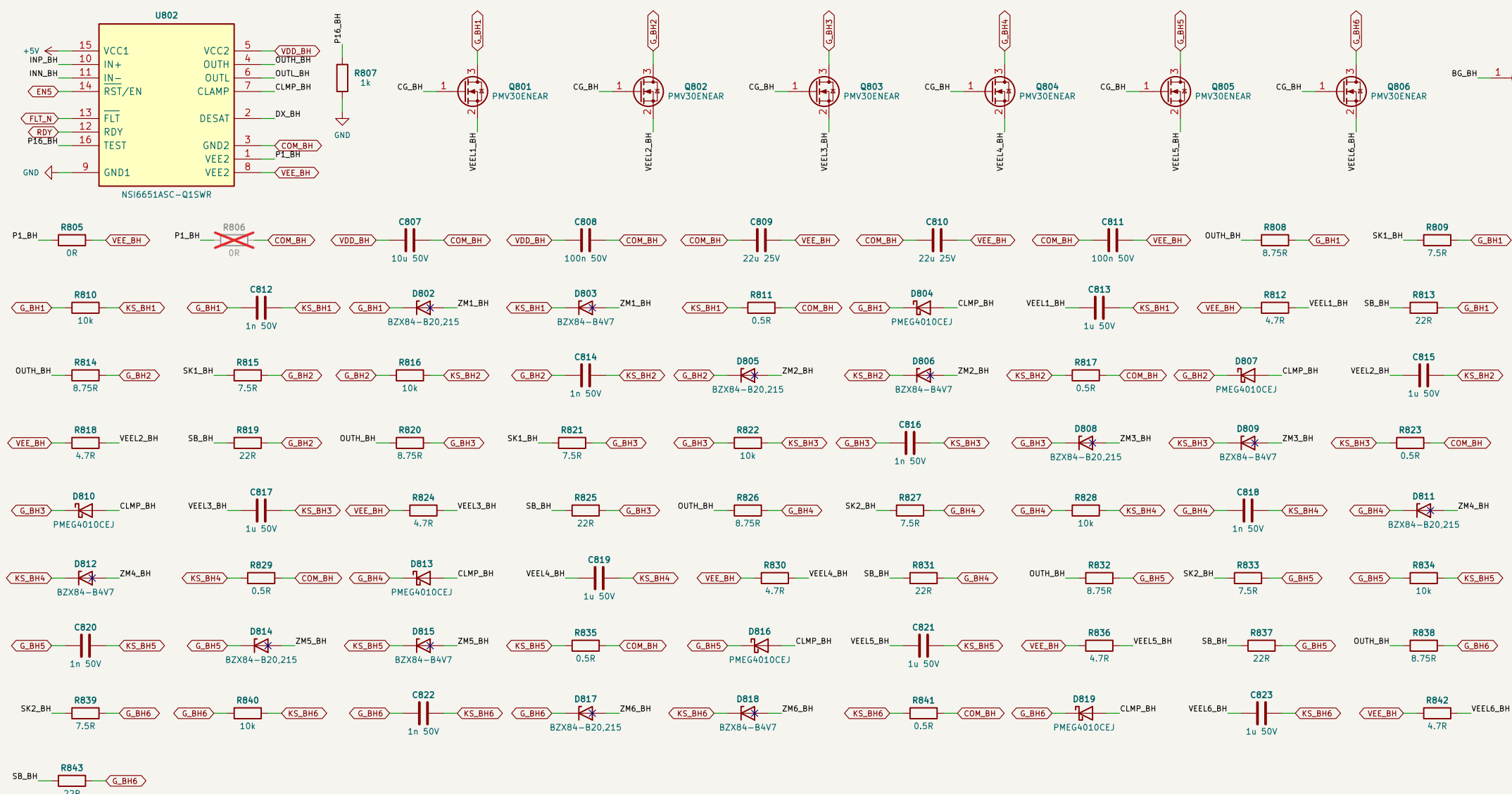
Gate and Kelvin of each device as coupled pairs of equal length from the COM star; Rdt: Kelvin; R: 10k ± 1%; Zeners and the clamp FET = C.A. at each point (Rdt = 500p, R = 500p); PNP followers next to their three gates; DESAT: wiring >= 2.5 mm sag gap per device, cool to VEE; Miller hold of AL gates NOT simulated - BENCH TEST.

BH: LOGIC SIDE



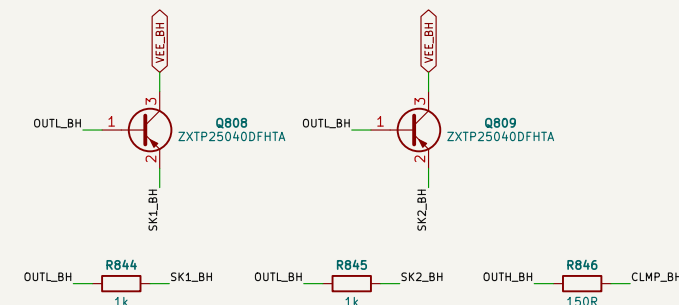
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BH: DRIVER AND 6 GATES



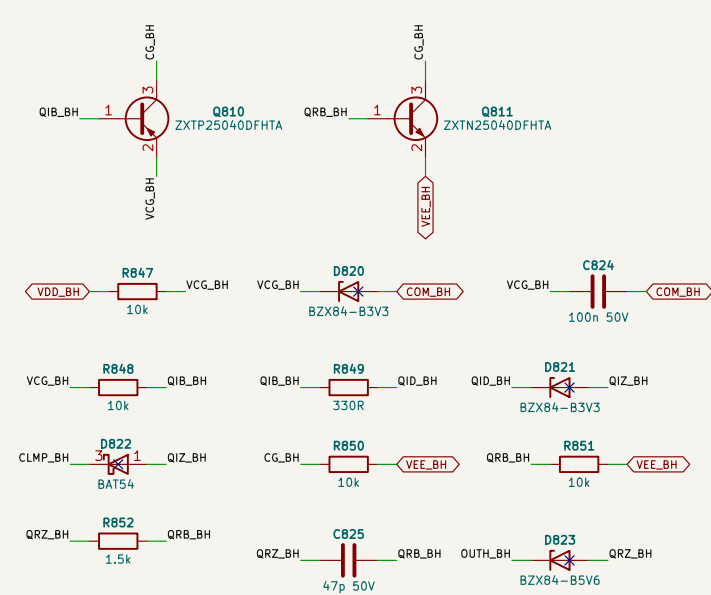
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Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

BH: TURN-OFF BUFFER



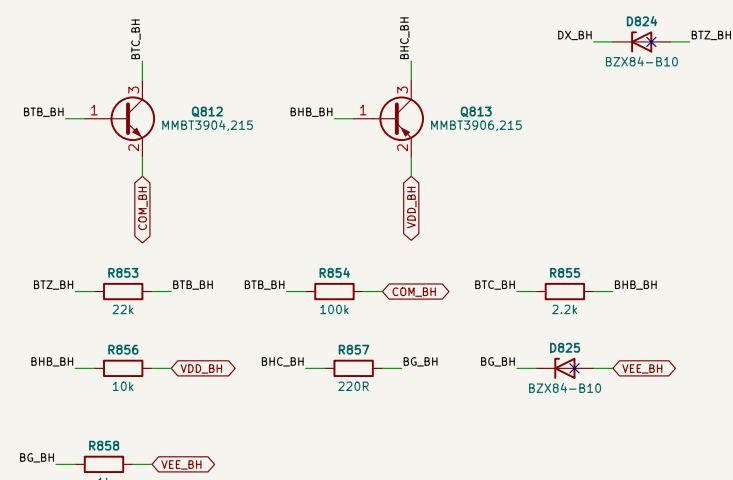
2 x XZPT25040DFH emitter followers on OUTL (B-E 1k, C to VEE): each sinks the Roff current of 3 gates; the NSI6651 supplies base current only. Turn-on stays on OUTH.

BH: MILLER CLAMP CONTROL



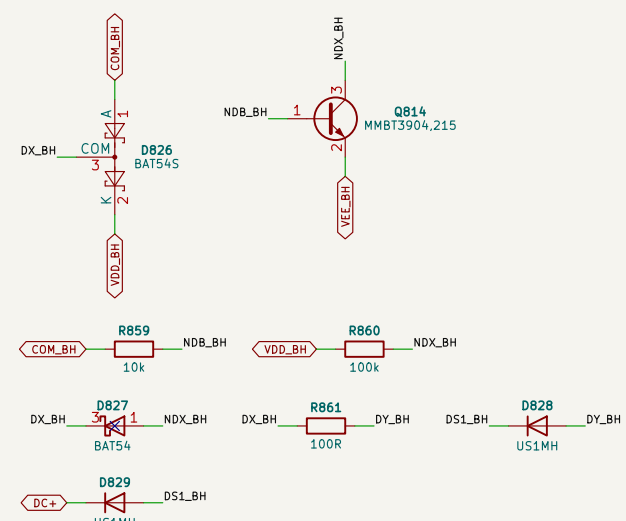
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

BH: SHORT-CIRCUIT BOOSTER



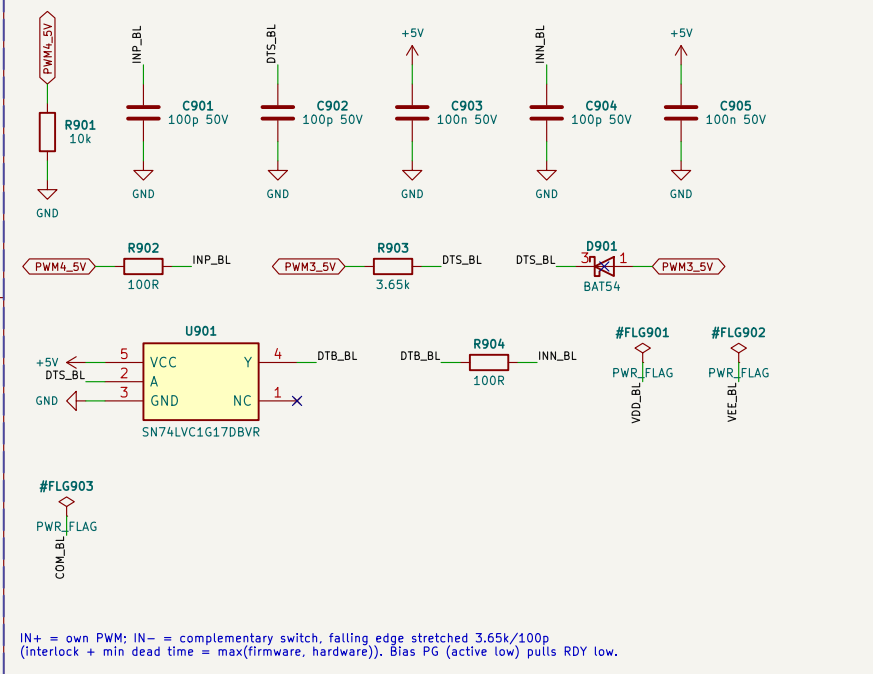
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

BH: DESAT

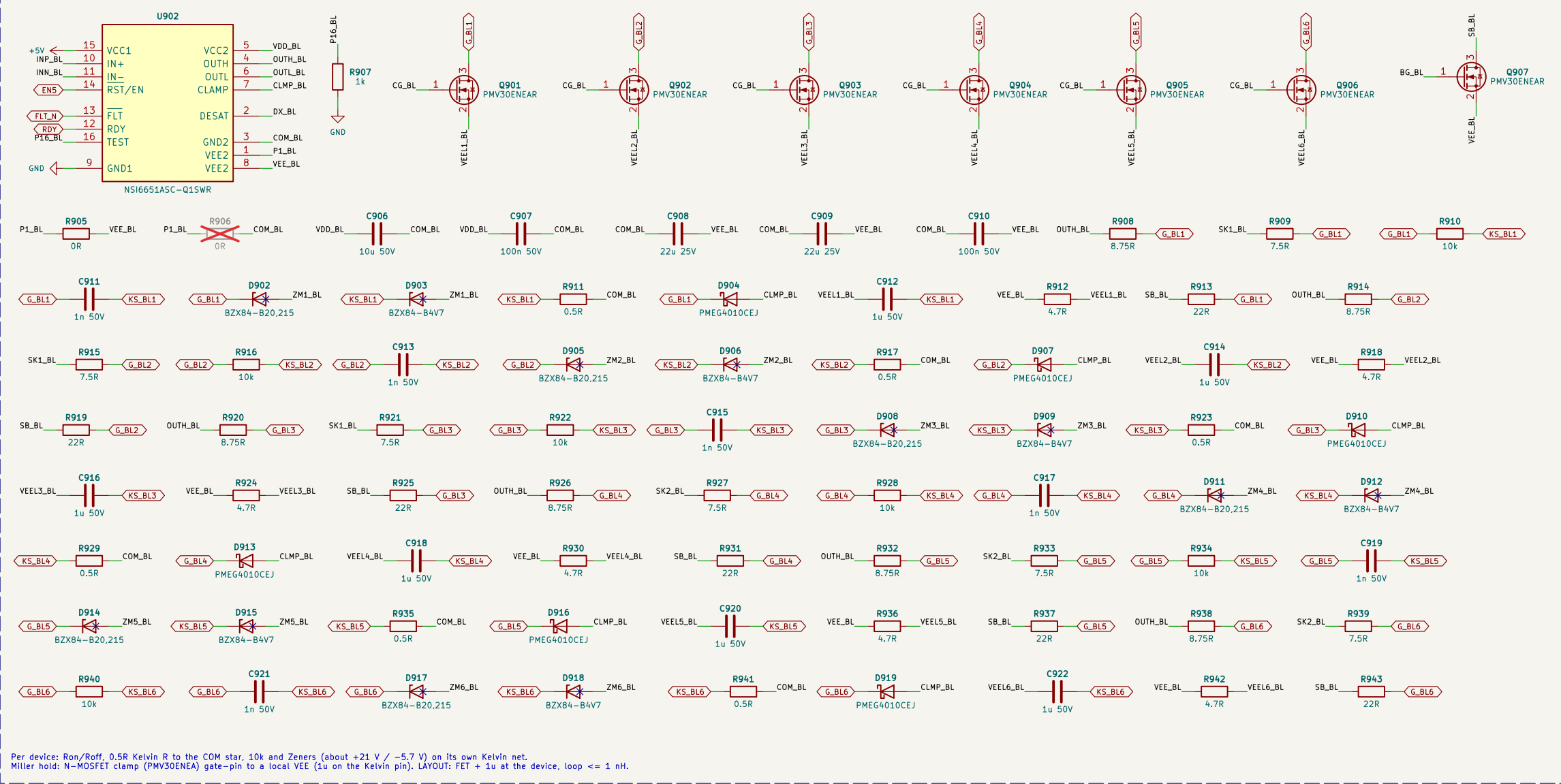


DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

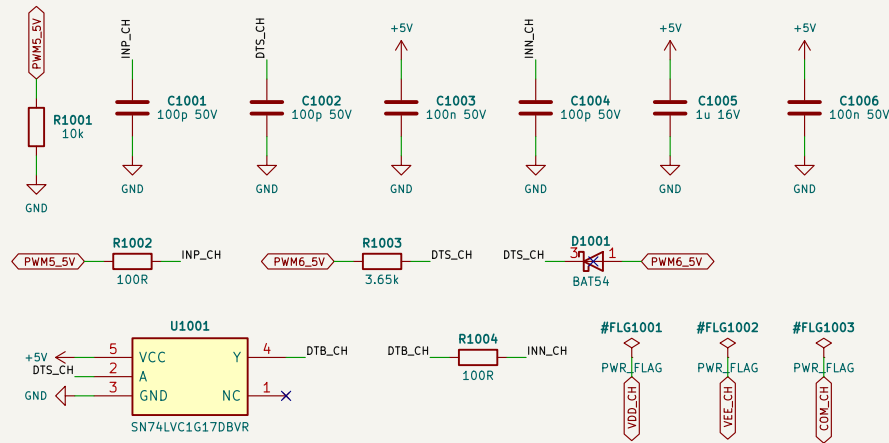
BL: LOGIC SIDE



BL: DRIVER AND 6 GATES

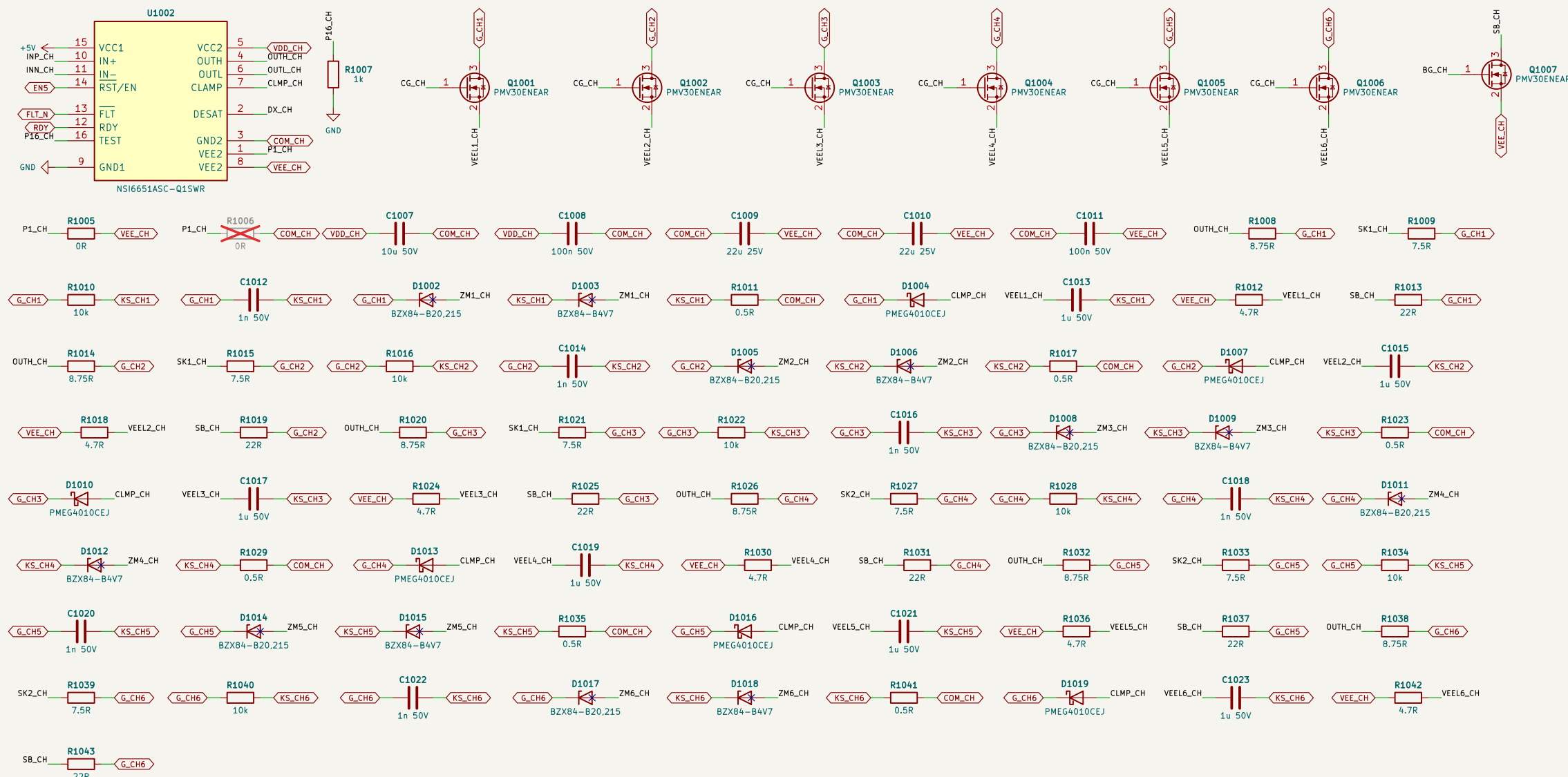


CH: LOGIC SIDE



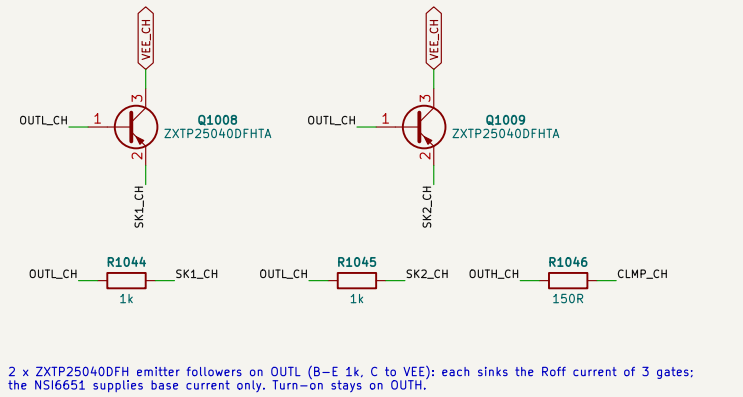
IN+ = own PWM; IN- = complementary switch, falling edge stretched 3.65k/100p
(interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

CH: DRIVER AND 6 GATES



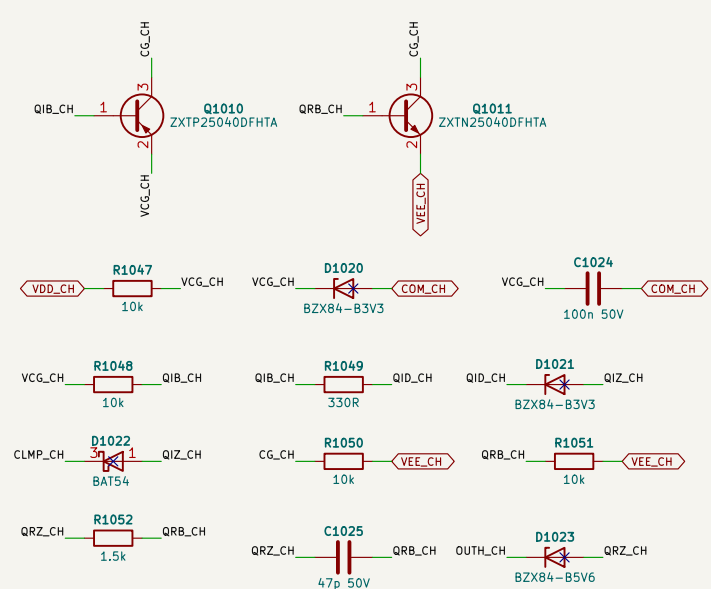
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin), LAYOUT: FET + 1u at the device, loop <= 1 nH.

CH: TURN-OFF BUFFER



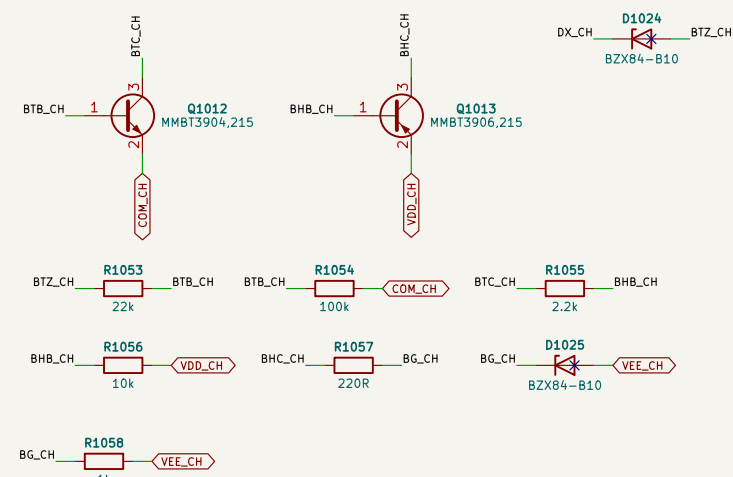
2 x ZXTF25040DFHTA emitter followers on OUTL (8-T 1k, C to VEE); each sinks the Roff current of 3 gates;
the NSI6651 supplies base current only. Turn-on stays on OUTH.

CH: MILLER CLAMP CONTROL



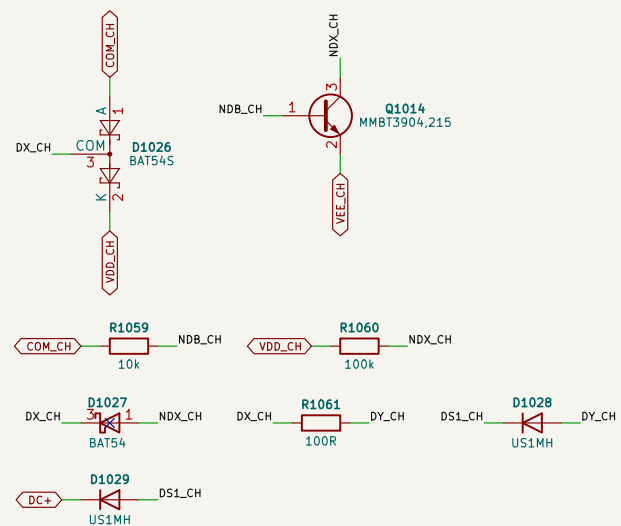
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

CH: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

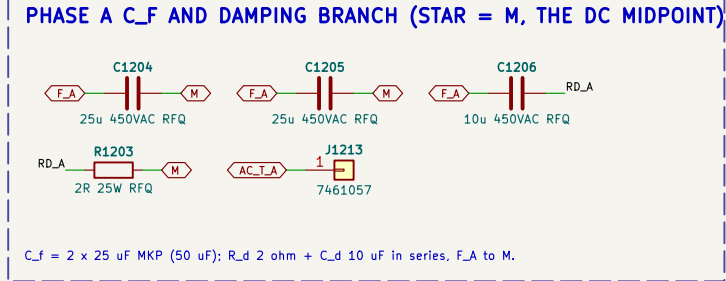
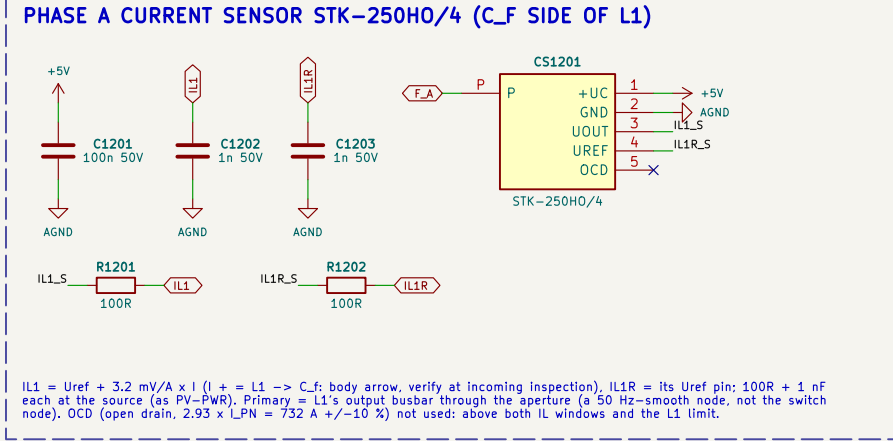
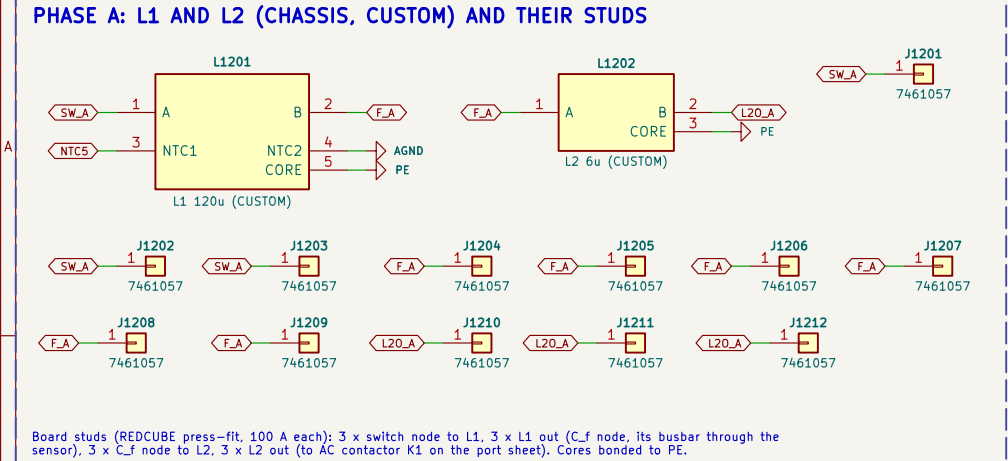
CH: DESAT

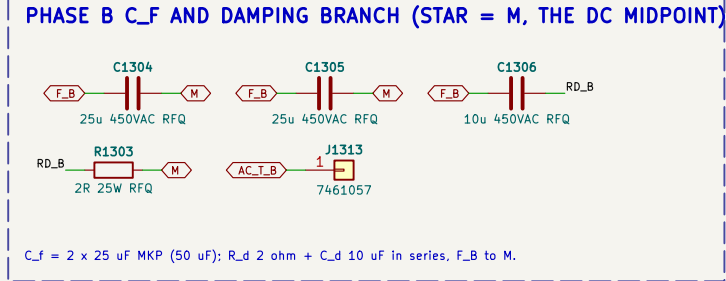
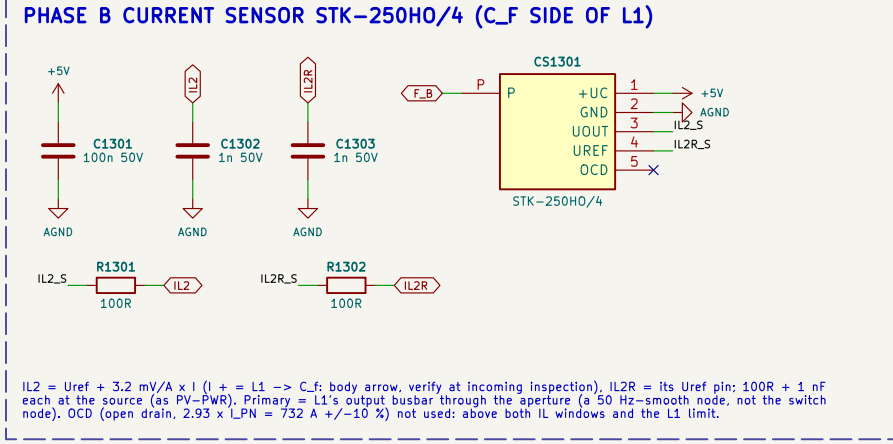
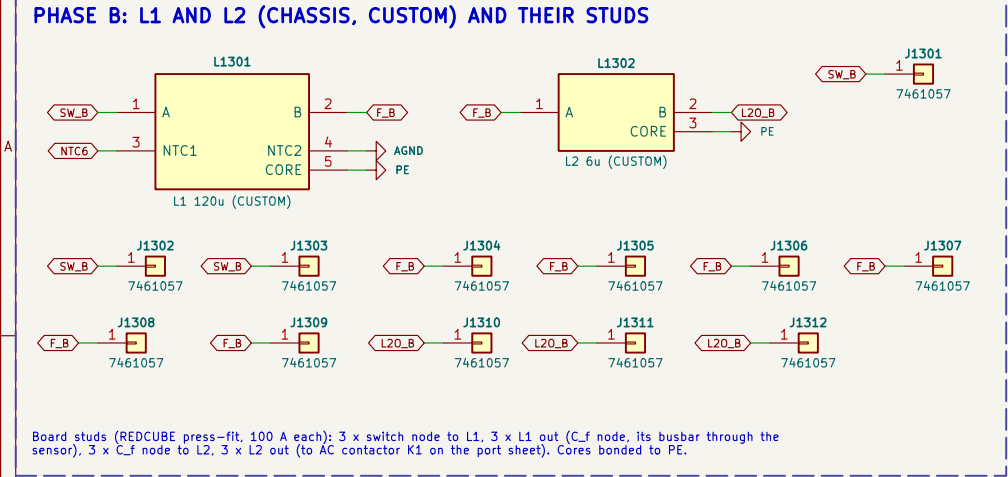


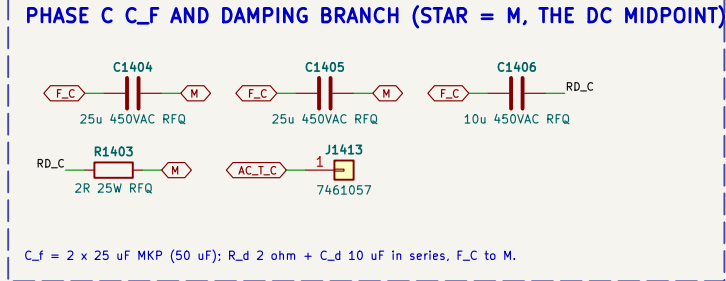
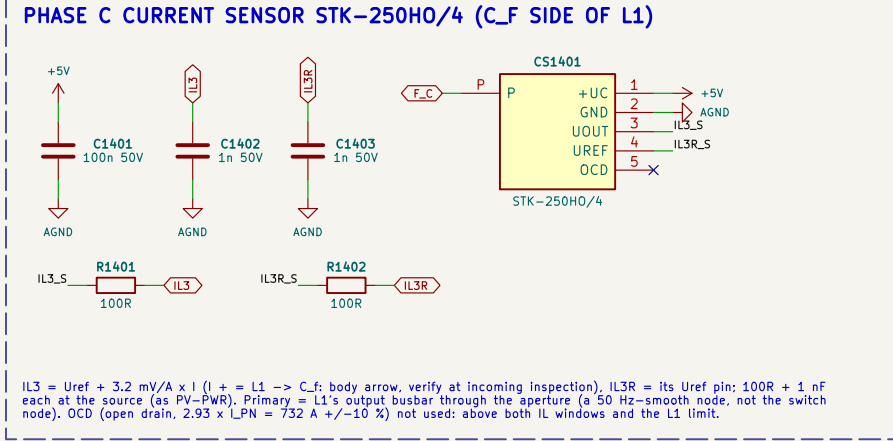
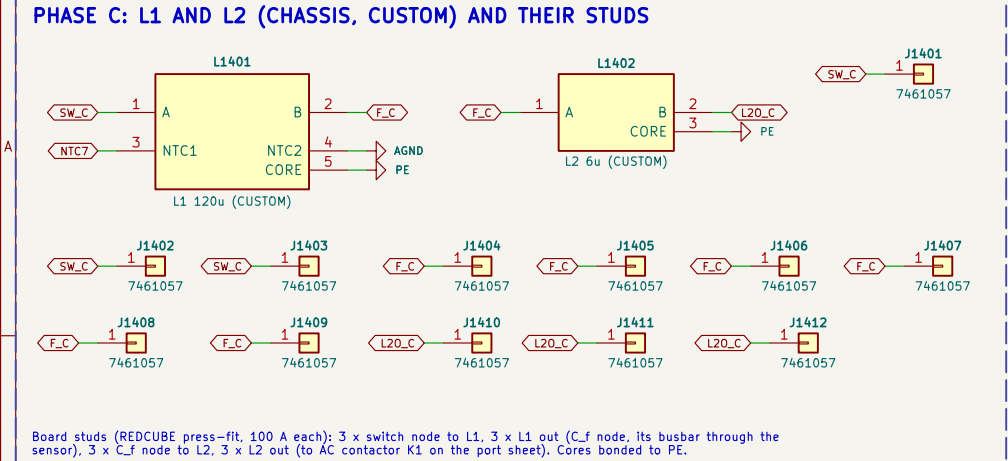
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - Rs ~ 2 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 11
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

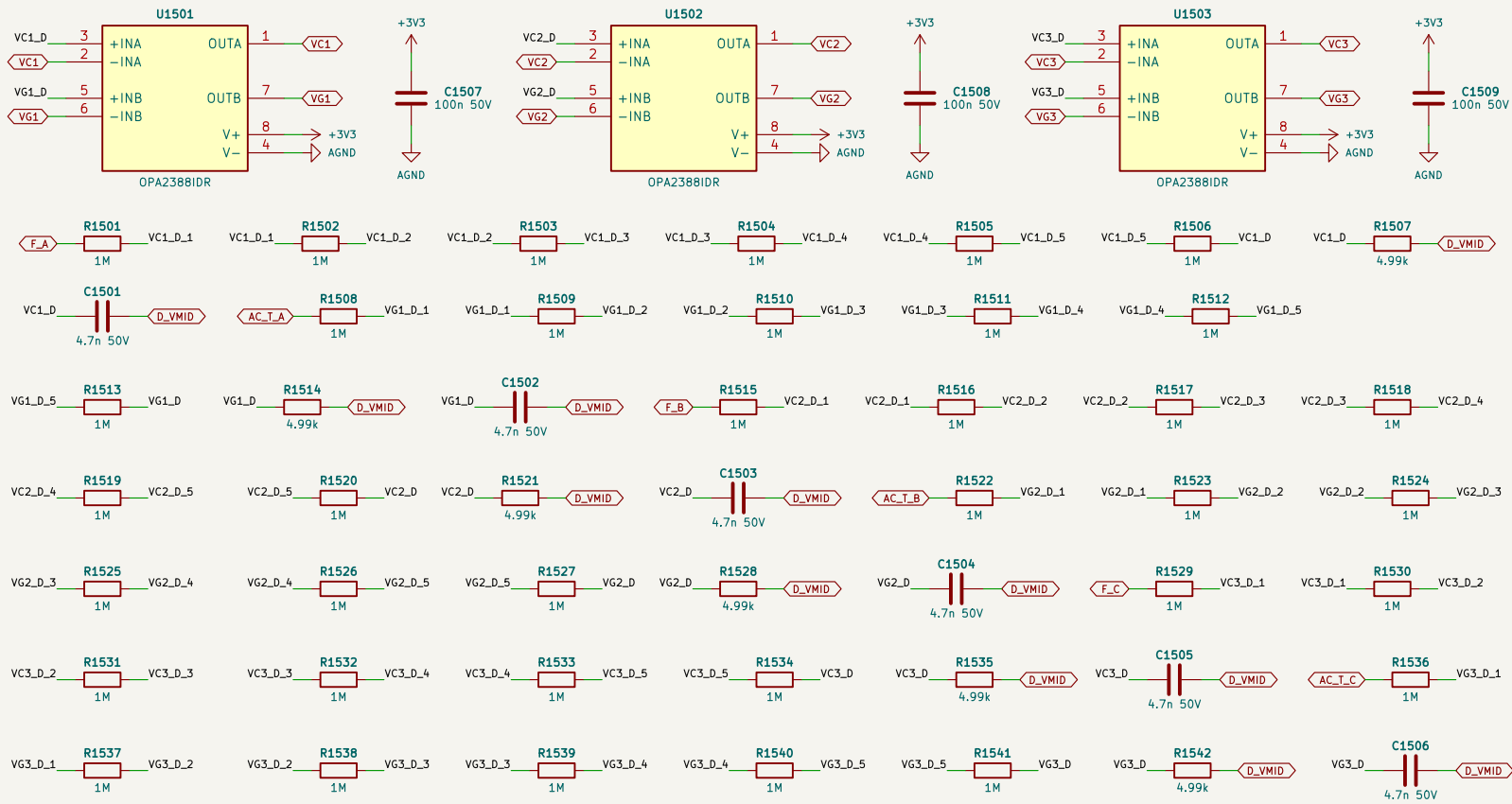
Sheet: /10 Leg c high-side gate drive/
File: 10_gdch.kicad_sch
Title: PCS-P125 power - Leg c high-side gate drive
Size: A2 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 11/27







DIVIDERS (1 V = 1203 V AROUND VMID, BIPOLAR: THE TERMINALS FLOAT AGAINST DC- WITH THE CONTACTORS OPEN)



Relay test: VG vs VC per phase with one contactor closed at a time (firmware).

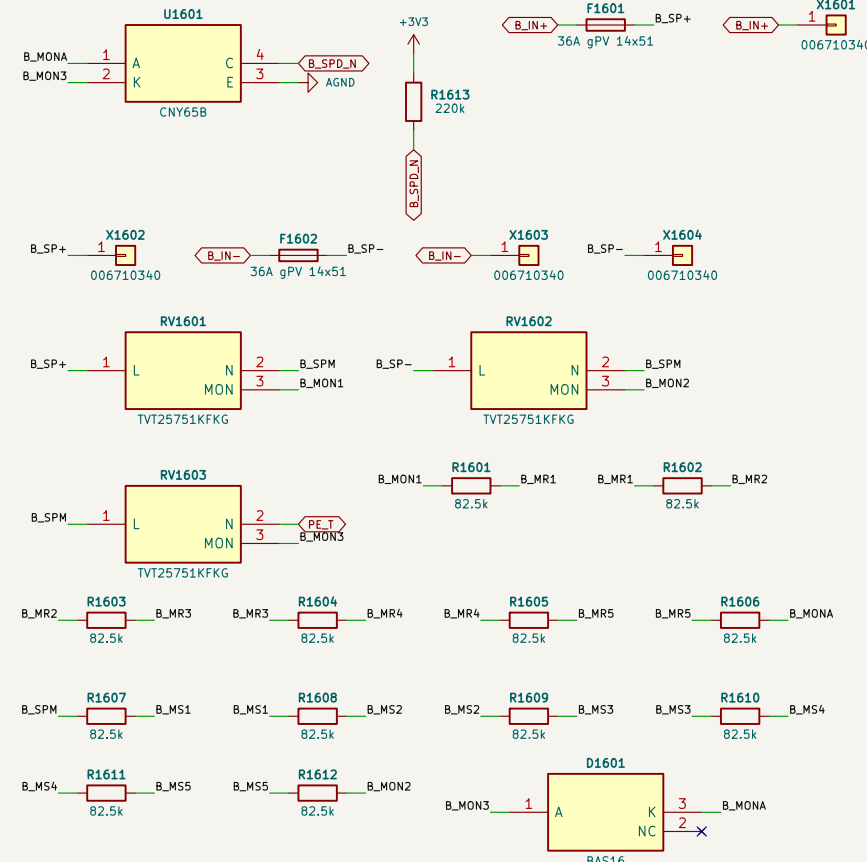
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /15 AC voltage sensing/ File: 15_acsense.kicad_sch		
Title: PCS-P125 power – AC voltage sensing		
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 16/27

PORT B (BATTERY SOURCE), D-044 LEAN PORT

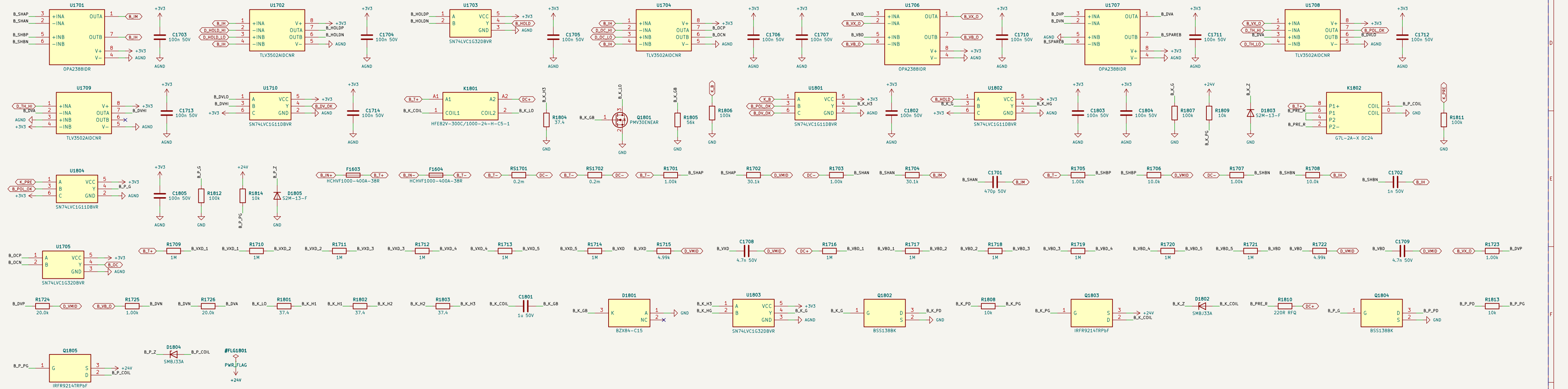
Terminal side B_{1N}+ / B_{1N}-, bank DC+. HCHVF1000-400A-38R aR in both poles.
Contactor opens <= 0.97 kA, held above (port_spec lean). Fuse / contactor coordination: the board's design check.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TVT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up,eff 3793 V pole-PE and pole-pole at In 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA, Branch fuses ETI CH14x51 gpV36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

AR FUSES (BATTERY-TYPE PORT)



HCHVF1000-400A-38R in both poles (aR, partial range): see the board's design check.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_checks()
 Sheet 17
 Schematics authored by Chinmoy Bhuyan
 Live board referenced to DC--; gate islands functional; HV-PE basic
 Sheet: /16 Port B power path (battery)/
 File: 16_leanB_power.kicad_sch
Title: PCS-P125 power – Port B power path (battery)
 Size: A2 Date: 2026-10-05 Rev: A0
 KiCad E.D.A. 10.0.6 Date: 17/27

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /17 Port B sensing and interlocks/
File: 17_leanB_ctrl.kicad_sch

Title: **PCS-P125 power – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: A0

KiCad E.D.A. 10.0.6

Id: 18/27

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 19
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /18 Port B coil drives/
File: 18_leanB_drive.kicad_sch

Title: PCS-P125 power – Port B coil drives

Size: A3

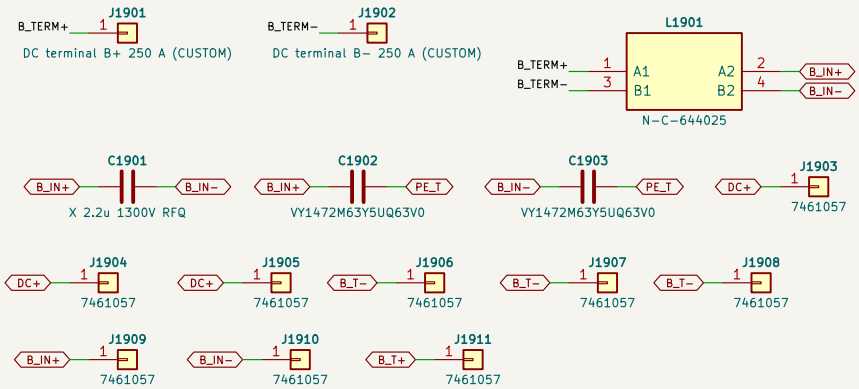
Date: 2026-10-05

Rev: A0

KiCad E.D.A. 10.0.6

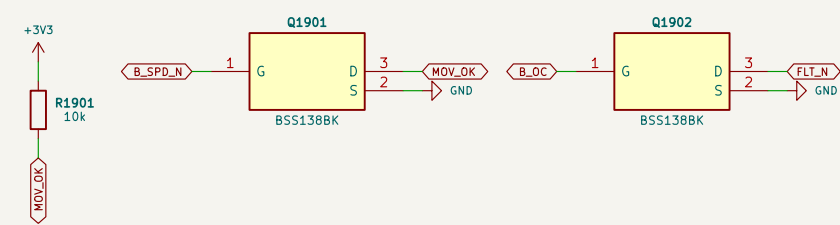
Id: 19/27

DC TERMINALS AND FILTER



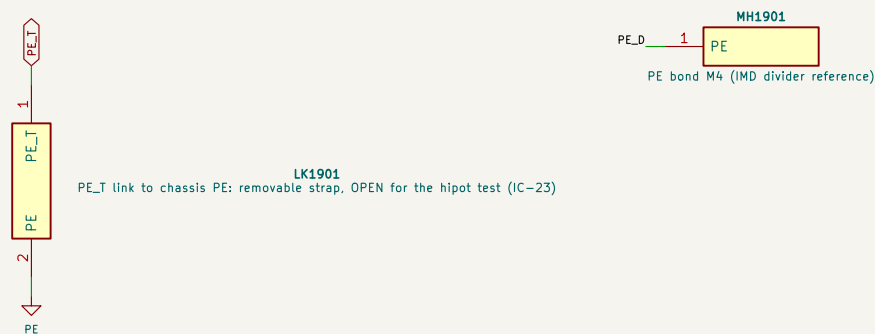
B_TERM+/- (250 A), X 2.2 uF 1300 V across the port, Y1 4.7 nF pole - PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- by short links.

PORT STATUS TO PCS_PC



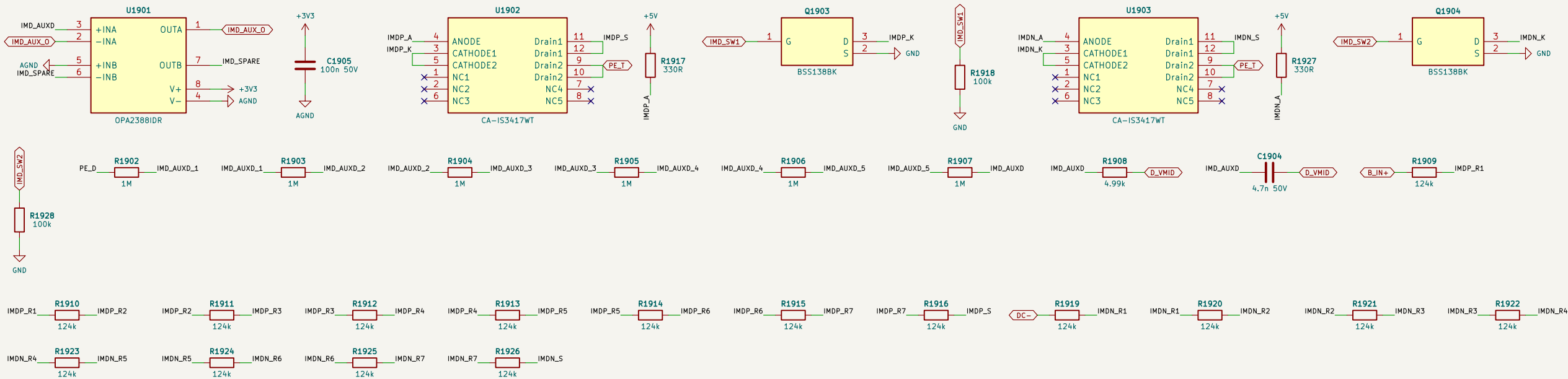
HOLD = B_HOLD (||| >= 0.97-1.03 kA: contactor held). MOV_OK high while the varistor monitor loop conducts. Port over-current (387-413 A) pulls FLT_N low (open drain); the latch trips without firmware.

PE



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot).

INSULATION MONITOR (PORT.LEAN_IMD)



Battery side, contactors open (IEC 62109-2 style check before connecting); IMD_SW1 = string B_IN+ -> PE, IMD_SW2 = PE -> DC-; VPE to PCS_PC.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()

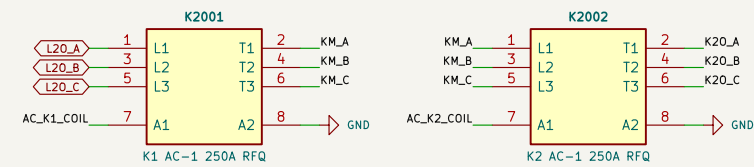
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /19 DC terminals, insulation test/
File: 19_dcterm.kicad_sch

Title: PCS-P125 power - DC terminals, insulation test

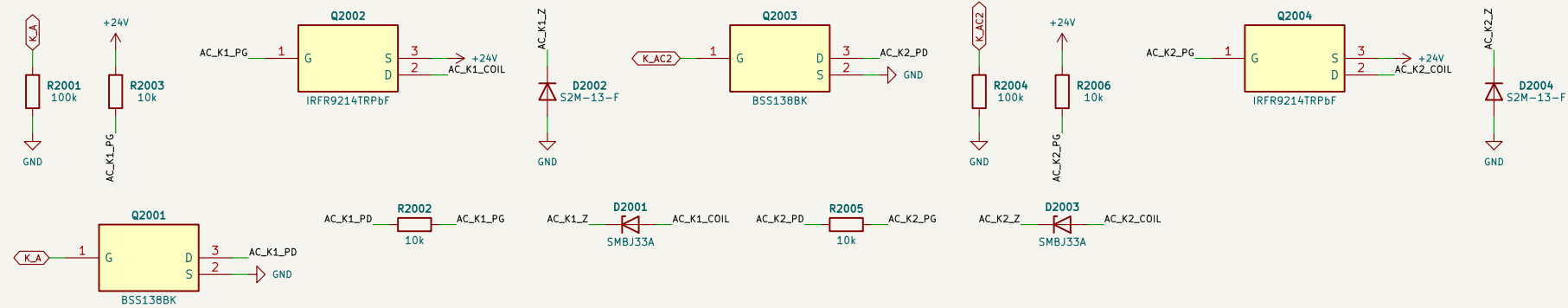
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 20/27

AC CONTACTORS K1, K2 (CHASSIS, RFQ): TWO 3–POLE CONTACTORS IN SERIES



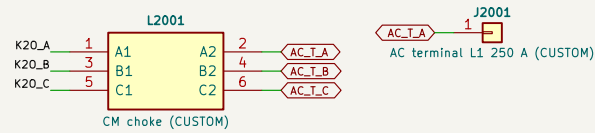
Relay test before every connection: K1 alone, then K2 alone; the voltage across the open one is $V_G - V_C$ (stage–3 dividers). Close only after synchronisation, open only at zero current (firmware).

COIL DRIVERS (PORT_LEAN_DRV): BSS138 → IRR9214 FROM THE LIVE 24 V, S2M + SMBJ33A CLAMP



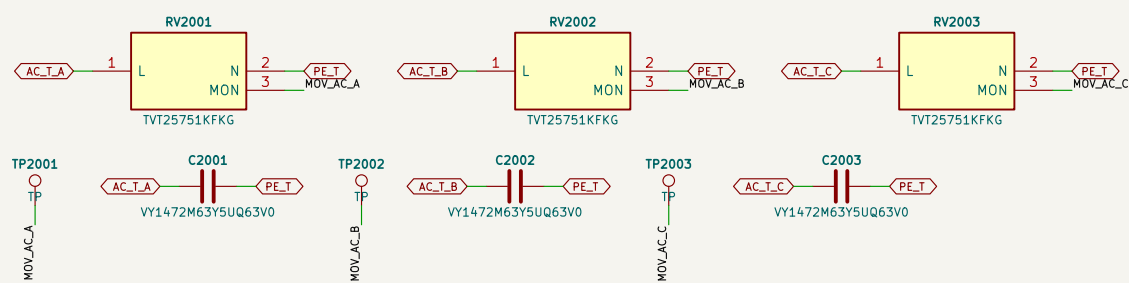
K1 from K_A (= K_AC1), K2 from K_AC2; logic lost = coil off. The contactors' DC coil modules economise.

CM CHoke (CHASSIS, CUSTOM) AND AC TERMINALS



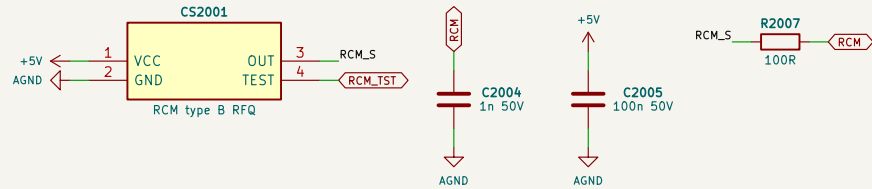
One turn of the three phase busbars through the cores; terminals L1–L3 250 A; PE terminal on the chassis.

AC SURGE PROTECTION (TYPE II) AND Y1



Per phase a thermally protected varistor TVT25751 (465 V AC) and a Y1 4.7 nF to PE_T; the removable PE link opens them for the hipot. Monitor leads to test points (OPEN 5).

RESIDUAL–CURRENT SENSOR (TYPE B, RFQ)



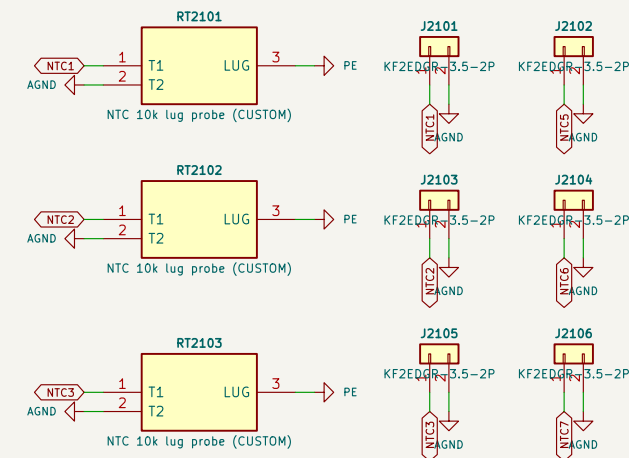
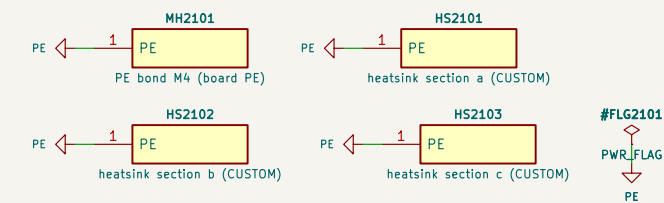
Around L1–L3 between the CM choke and the terminals; RCM analog out and RCM_TST to PCS.X.

Not bench–validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 21
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC–; gate islands functional; HV–PE basic

Sheet: /20 AC port: contactors, SPD, RCM/
File: 20_acport.kicad_sch

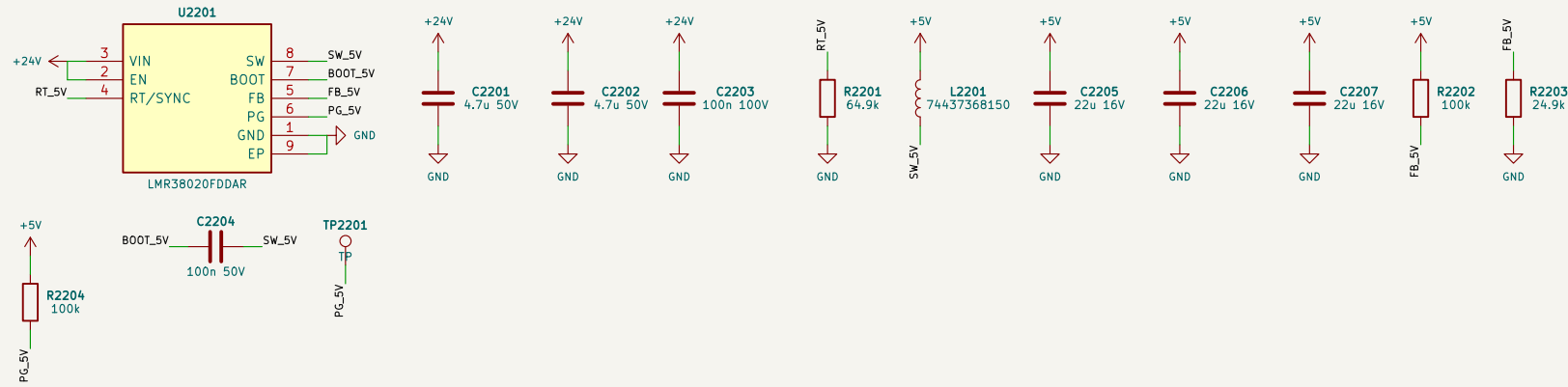
Title: PCS–P125 power – AC port: contactors, SPD, RCM

Size: A3 Date: 2026–10–05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 21/27



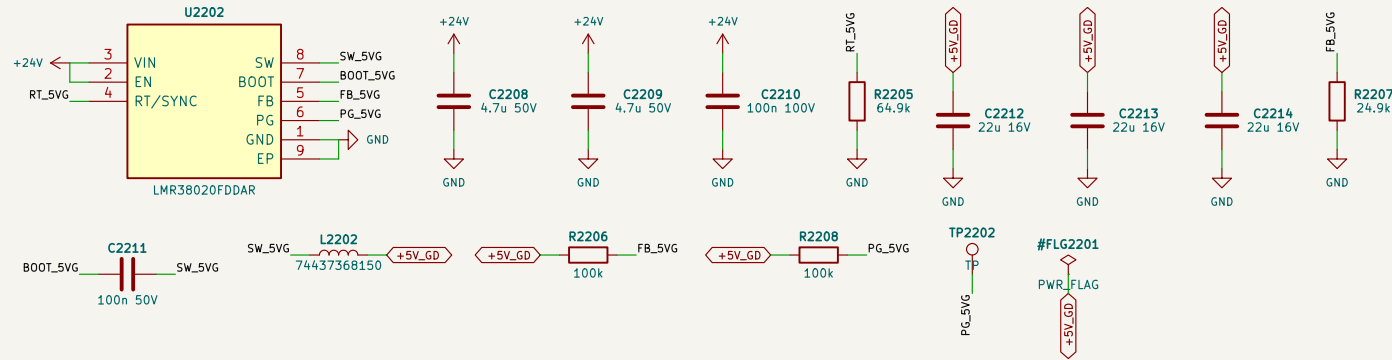
NTC1-3 = heatsink sections a-c: lug probe on the EARTHED heatsink, ≥ 2.5 kV rms lead-to-lug (R-10).
NTC5-7 = L1 winding NTCs (leads on 2-pin headers), 100 nF at each header.

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



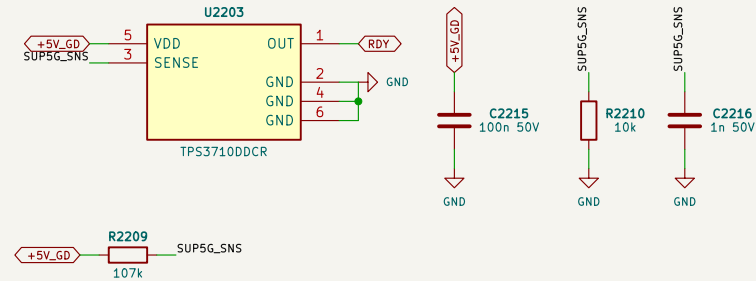
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



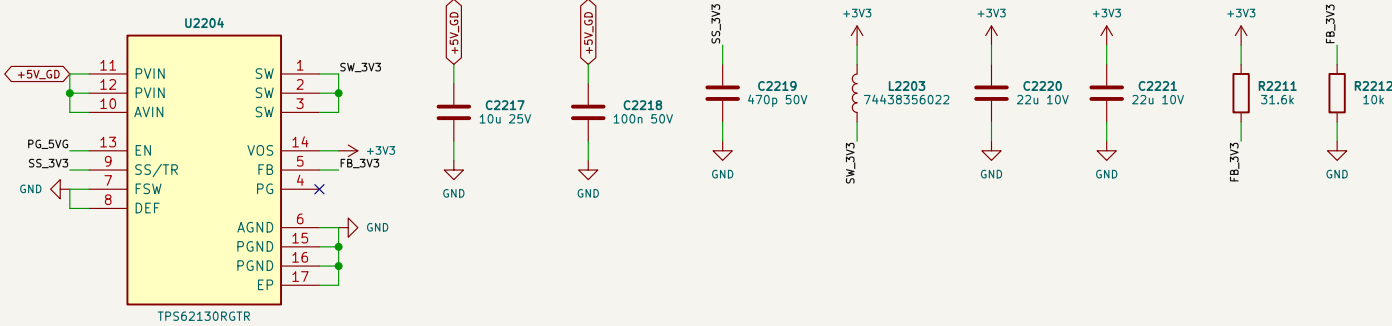
gate bias of legs b and c, +3V3 (TPS62130), ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



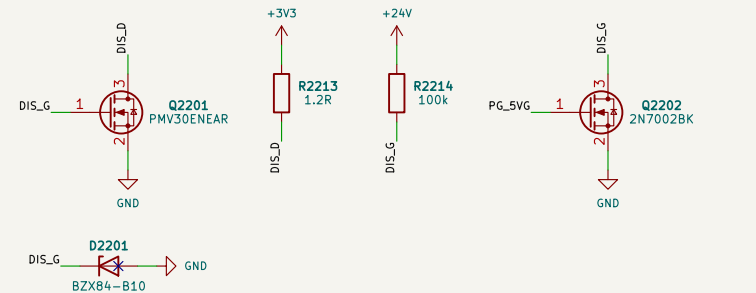
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us), Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

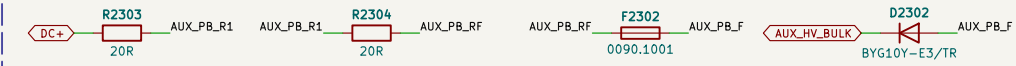
Sheet: /22 Live 5 V / 3.3 V, SELV connector/
File: 22_supplies.kicad_sch
Title: PCS-P125 power - Live 5 V / 3.3 V, SELV connector
Size: A3 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 23/27

TAP B_T+



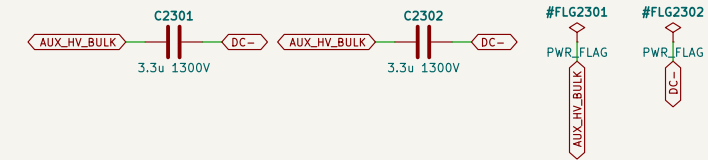
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current <= 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP DC+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

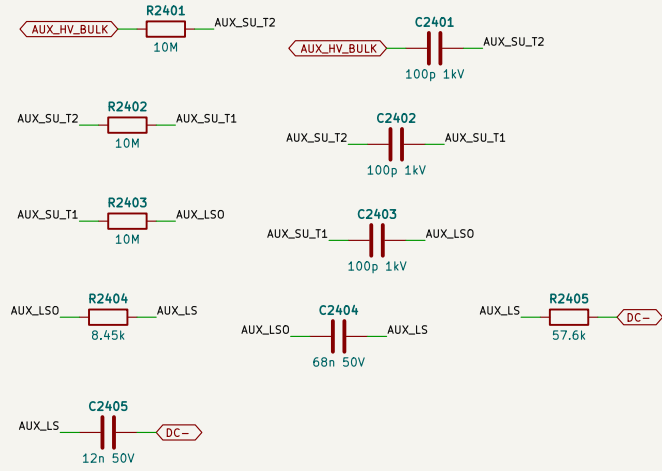
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 24
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /23 AUX75 input taps, OR, film/
File: 23_aux75_input.kicad_sch

Title: PCS-P125 power - AUX75 input taps, OR, film

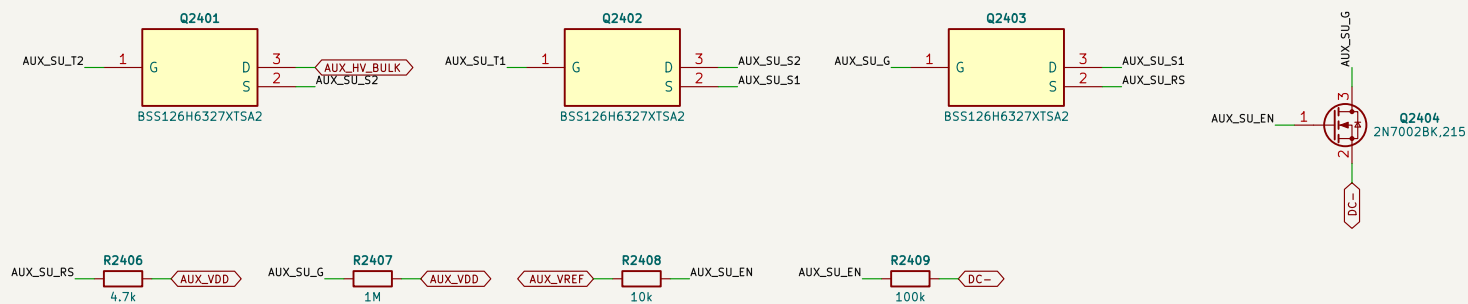
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 24/27

COMPENSATED TAP STRING



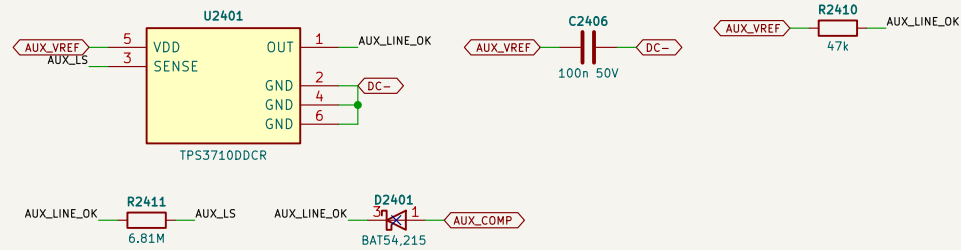
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



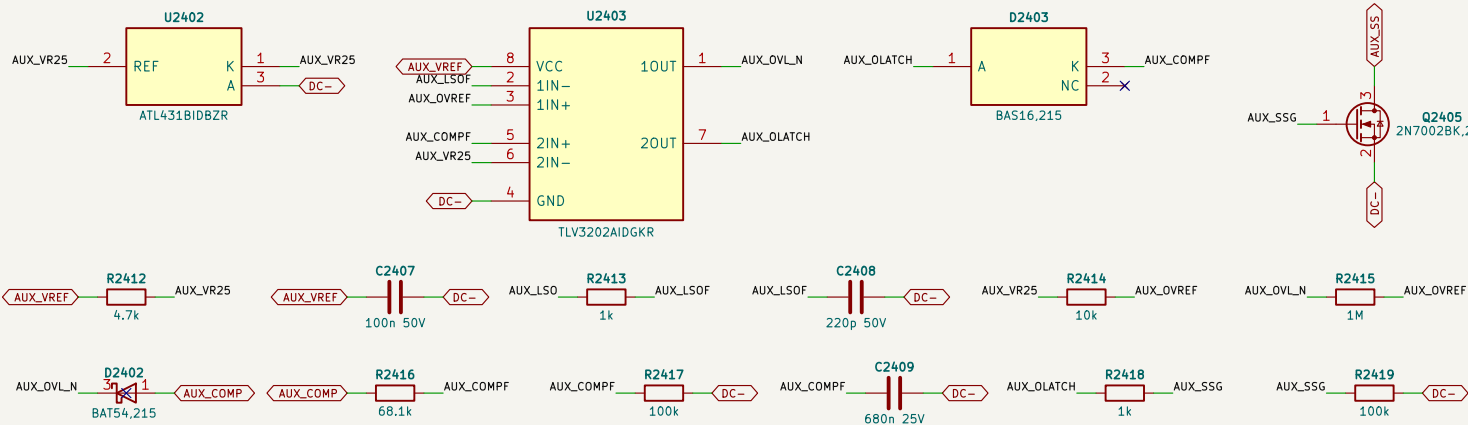
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

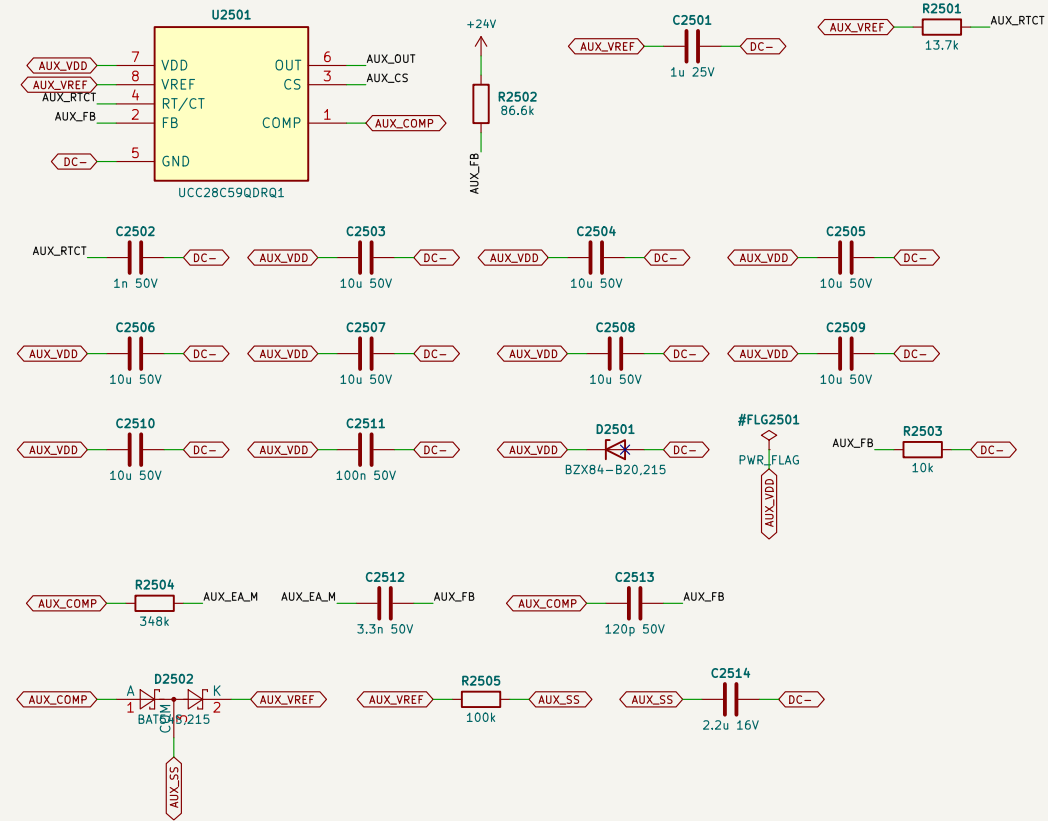
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 25
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /24 AUX75 start-up, brown-in, OV, timer/
File: 24_aux75_startup.kicad_sch

Title: PCS-P125 power – AUX75 start-up, brown-in, OV, timer

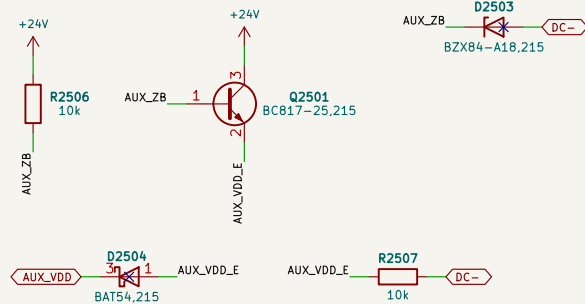
Size: A3 Date: 2026-10-05 Rev: A0
KiCad E.D.A. 10.0.6 Id: 25/27

PWM CONTROLLER AND ERROR AMPLIFIER



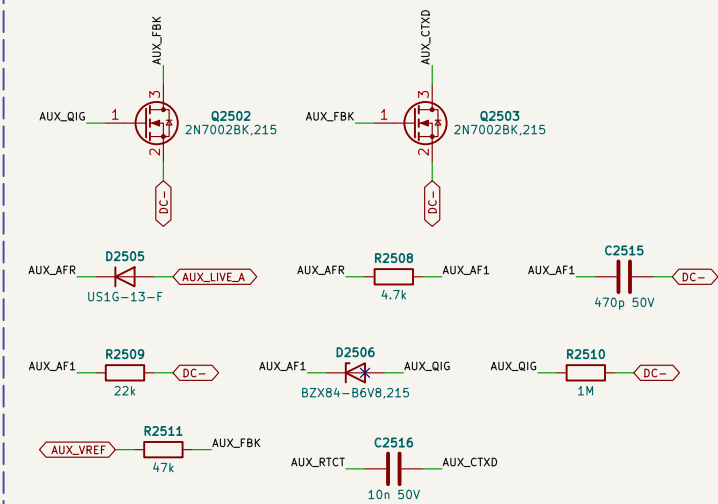
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



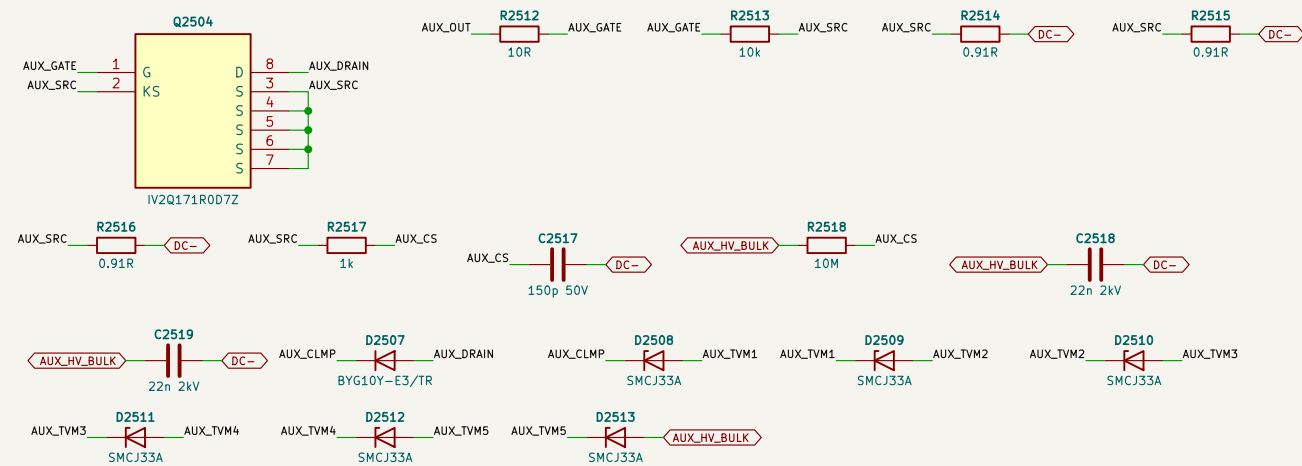
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 °C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



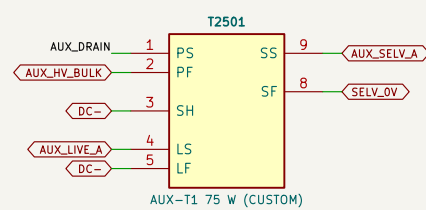
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 In parallel = 0.303333R; lowest limit 2.90 A delivers 103 W (>= 110 % of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK -> CS (CRH2512); CS offset 101 mV at 1000 V cancels most of the 155-250 nS junction overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ33; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/- 7 %; shield and live winding on DC-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

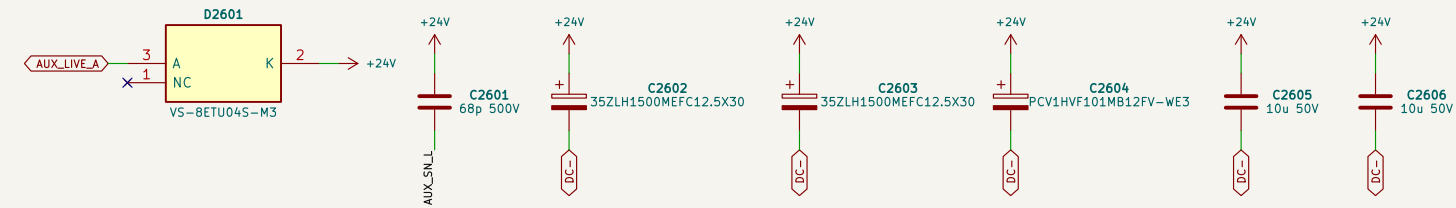
Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 26
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /25 AUX75 PWM, SiC, clamp, AUX-T1/
File: 25_aux75_power.kicad_sch

Title: PCS-P125 power – AUX75 PWM, SiC, clamp, AUX-T1

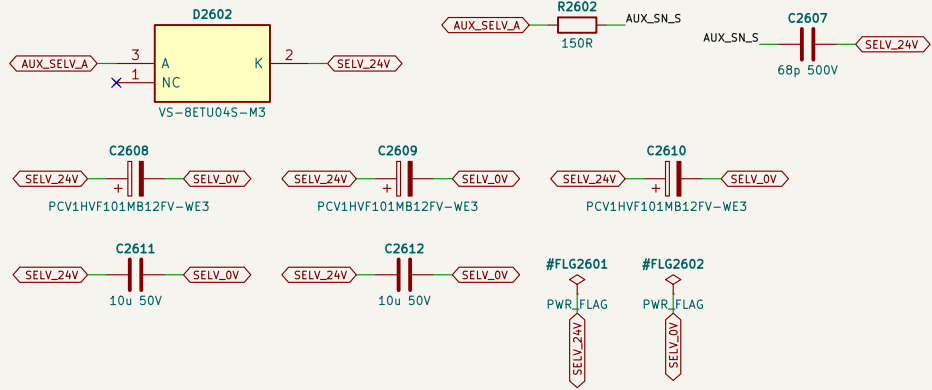
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 26/27

LIVE 24 V RECTIFIER, HOLD-UP



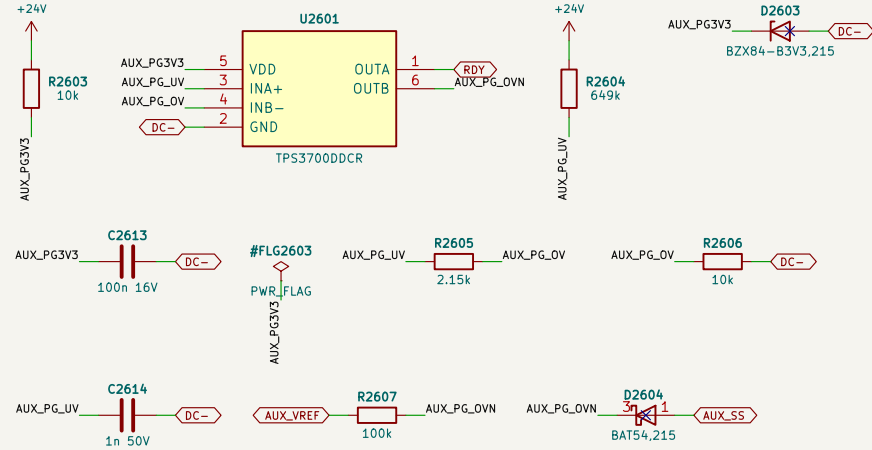
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)

Not bench-validated. Values read from sim/out/pcs_design/pcs_spec.json; checks: design_check()
Sheet 27
Schematics authored by Chinmoy Bhuyan
Live board referenced to DC-; gate islands functional; HV-PE basic

Sheet: /26 AUX75 live 24 V, SELV 24 V, status/ File: 26_aux75_outputs.kicad_sch		
Title: PCS-P125 power – AUX75 live 24 V, SELV 24 V, status		
Size: A3	Date: 2026-10-05	Rev: A0
KiCad E.D.A. 10.0.6		Id: 27/27